ADBMS6830 Preliminary Safety Manual
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SCOPE

The ADBMS683x family of Multicell Battery Monitors includes multiple derivative devices. The scope of this document is limited to ADBMS6830 derivatives with the 'W' and the 'FS' in the part name, branding them as subject to automotive quality and functional safety processes. The ADBMS6830WFS devices are developed for use in ISO 26262 applications for Automotive Safety Integrity Level Capability D (ASIL D).

According to the 2018 version of the ISO 26262 "Road vehicle – Functional Safety" standard, part 10, cause 9, the ADBMS6830 series is developed as a Safety Element out of Context (SEooC) to be used in multiple automotive safety applications. Because of the SEooC approach, this document does not intend to present exhaustive hazard and risk analysis of the system into which the ADBMS6830 is integrated, but to document the assumptions Analog Devices made when addressing expected requirements for battery monitoring systems (BMS). During concept and system phases, Analog Devices have made assumptions about the item and the target application, representative use cases and resulting safety requirements in line with SEooC guidance, and verifies the device's capability to meet these assumed safety requirements with a given ASIL at the device level. It is the responsibility of the system integrators to validate whether the assumed requirements fulfill the needs of a concrete application case. The ADBMS6830 devices are not restricted to be used in scenarios with diverting safety requirements, as long as the system integrators verify the safety of the parent system.

The objective of this safety documentation is to provide the necessary guidance for the integration of ADBMS6830WFS into safety-critical systems when showing compliance with the ISO 26262-2018 functional safety standard. This version of the document is not intended to guide on compliance with the IEC 61508 and ISO 21448 safety standards.

PRODUCT OVERVIEW

The ADBMS6830 series is a family of multicell battery stack monitoring devices that measures in series connected battery cells. The measurement input range of -2.5V to 5.5V makes the ADBMS6830 suitable for most battery chemistries and it also allows to measure voltages across bus bars. All connected cells can be measured simultaneously with individual ADCs. The ADCs have a sampling rate of about 2 MHz and a conversion time of 1 ms. Higher noise reduction can be achieved by subsequent CAVG8 (FIR) filters and programmable IIR filters. Additional general-purpose inputs enable monitoring of cells temperature, measured as a voltage drop across a thermistor.

Multiple ADBMS6830 devices can be connected in series, permitting simultaneous cell monitoring of long, high voltage battery strings. Each ADBMS6830 has an isoSPI interface for high speed, RF-immune, long distance communications. Multiple devices are connected in a daisy-chain with one host processor connection for all devices. The device features bidirectional data transmission to protect against a broken wire.

The ADBMS6830 can be powered directly from the battery stack or from an isolated supply. The ADBMS6830 features an onboard 5V regulator. A hierarchy block diagram of ADBMS6830 with the main functional safety related blocks is depicted in Figure 1.

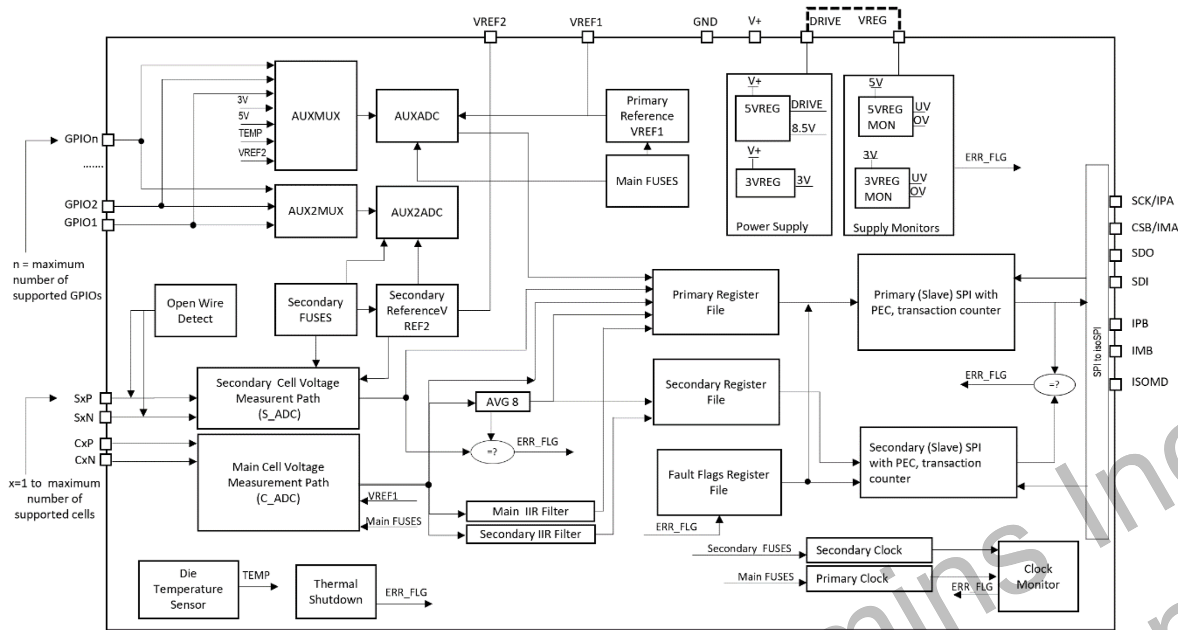

Figure 1. ADBMS6830 Conceptual Diagram

Analog Devices offers further products complementary to the ADBMS6830 series:

- ADBMS682x isoSPI Communication Devices

Although this document may cross reference to these devices, the safety concept of the ADBMS6830 series stands by its own.

Each device is programmed at the factory with a unique 48-bit serial identification code (SID) which is stored in the SID Register. The SID code for each device can be read using the RDSID command. The device type identifier (DEVID) contained in the SID is set to 0x03 for ADBMS6830 derivatives. The revision code (REV) is returned to the RDSTATE command. For current Revision C silicon the RDSTATE returns REV = 0x3. References specific to this silicon revision are labeled by "REV3" in the scope of this document. For other revision codes this document may need update.

TABLE OF CONTENTS

REVISION HISTORY

PrG 03/2022
Updated SM_SLEEP_IND;
Updated SM_NVM_ECC;
Added alternative SM_FREEZE_SEQ;
Updated bit descriptions;
Fixed VDEL and VDE bit positions;
Fixed COMP bit typo in SM_VCELL_CMP_DIAG;
Fixed AoU24 enumerator.

PrF 01/2022
Updated software sequences;
Updated V- open mitigation;
Added assumptions of use.

PrE 10/2021
Changes to SM_POWER_OOR;
Addition of SM_POWER_OOR_DIAG;
Reworded Open-wire sections;
Modified AoUs;
Optimization of sequences;
Various Clarifications.

PrD 02/2021
HW Metrics based on TARGET Safety Mechanisms assumed diagnostic coverages.

PrC 09/2020 (Rev 0.3)
Document reformatted.
Updated Safety Mechanism Names. See Table 4.
Add detailed description of Safety Mechanisms
Update diagrams
Update AoUs

Rev 0.2 | 01/2020
General rework

Rev 0.1 | 10/2019
Initial document

TERMINOLOGY

A

ADC
Analog to Digital Converter

ADI
Analog Devices Inc.

AoU
Assumption of Use

ASIL
Automotive Safety Integrity Level

B

BMC
Battery Management Controller

BMS
Battery Management System

C

CAN
Controller Area Network

CRC
Cyclic Redundancy Check

D

DFA
Dependent Failure Analysis

DTTI
Diagnostic Test Time Interval

E

EV
Electric Vehicle

F

FDTI
Fault Detection Time Interval

FIT
Failures In Time

FHTI
Fault Handling Time Interval

FM
Failure Mechanism/Mode

FMEDA
Failure Mode Effect and Diagnostic Analysis

FRTI
Fault Reaction Time Interval

FTA
Fault Tree Analysis

FTTI
Fault Tolerant Time Interval

G

GPIO
General Purpose Input/Output

H

HEV
Hybrid Electric Vehicle

I

IIR
Infinite Impulse Response (Filter)

L

LF
Latent Fault

LFM
Latent Fault Metric

LPCM
Low-Power Cell Monitoring

M

MPFDTI
Multiple-Point Fault Detection Time Interval

N

NDA
Non Disclosure Agreement

NTC
Negative Temperature Coefficient

NVM
Non-volatile Memory

O

OTP
One Time Programmable

OV
Overvoltage

P

PCB
Printed Circuit Board

PEC
Packet Error Code

PMHF
Probabilistic Metric for random Hardware Failures

PWM
Pulse Width Modulation

R

RF
Radio Frequency

S

SEooC
Safety Element out of Context

SM

Safety Measure/Mechanism

SPF
Single-Point Fault

SPFM
Single-Point Fault Metric

SPI
Serial Peripheral Interface

T

TME
Total Measurement Error

U

UV
Undervoltage

REFERENCE DOCUMENTS

ISO26262 Road Vehicles – Functional Safety, 2nd edition, 2018

ADBMS6830 Multicell Battery Monitor Data Sheet, Rev SpA, Dec 2021

ADBMS6830 Errata (currently no device anomalies are known and the errata sheet is clear).

ADBMS6830 IC Diagnostic Timing Sheet, January 2022

TARGET APPLICATIONS

ASSUMED TARGET APPLICATIONS

The safety concept assumes that the ADBMS6830 devices are deployed in Electric or Hybrid Electric Vehicles, which are subject to the definition of Road Vehicles as per the ISO26262:2018 standard.

The SEooC defines the assumed item as being the Electrical Storage System, consisting of a stack of the Battery Cells, Battery Management electronics, electrical and thermal sensors, actuators and contactors.

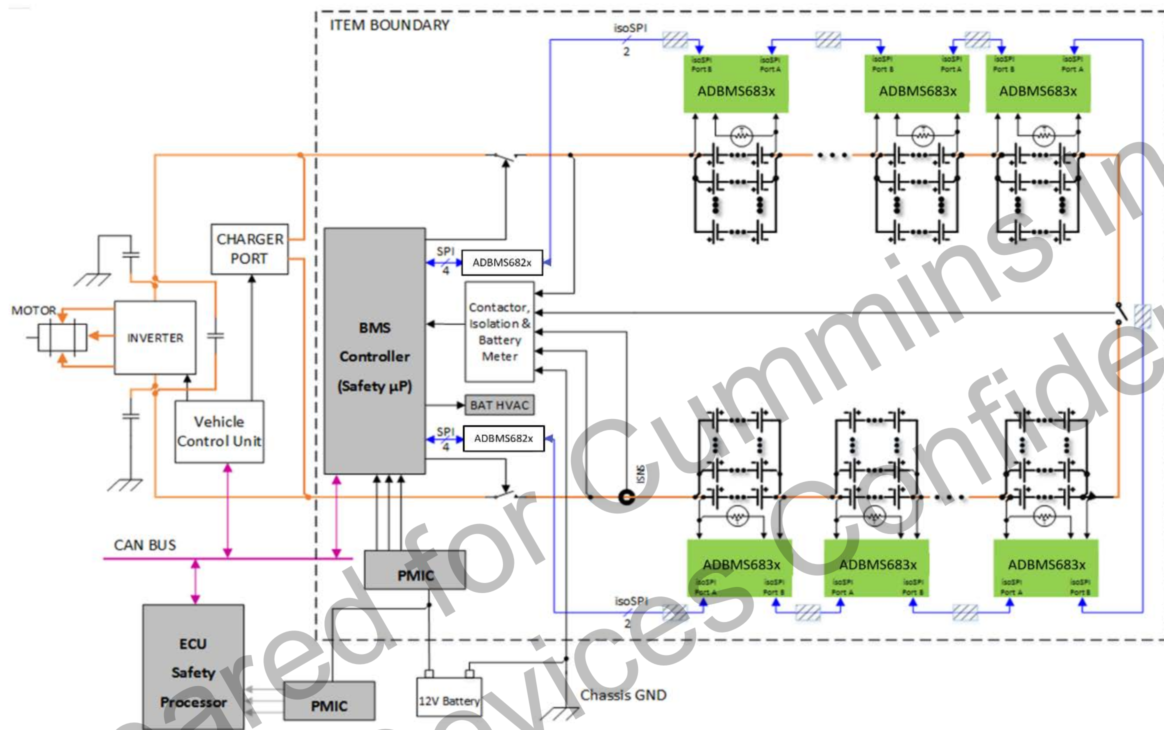

Figure 2. Item Definition

TYPICAL SYSTEM ARCHITECTURE

The SEooC assumes that the ADBMS6830 functions as a sensor device. It monitors the battery cells. However, the ADBMS6830 itself is supervised by a separate microcontroller, referred to as the BMS Controller (BMC) as shown in Figure 2. The BMC manages the cell monitoring measurements, performs the algorithms required to manage the Battery Cells and communicates with other items in the vehicle. The BMC also manages safety, if for any reason the ADBMS6830 was powered down or was not functional at all, and would no longer respond to commands from the BMC.

SAFETY REQUIREMENTS

ASSUMED FUNCTIONAL SAFETY REQUIREMENTS

The ADBMS6830 is a safety element out of context. Without knowing the system's safety goal, an assumption must be made regarding the role of the ADBMS6830 in achieving the system's safety goal. A Battery Management System (BMS) is the assumed top-level item which defines the safety goal(s).

To achieve the safety requirements, it is important to timely understand whether a fault in the cell voltage or cell temperature measurements has occurred. The assumed functional safety requirements are described in the Table 1.

The BMS controller must be responsible for putting the system to safe state, by reading and interpreting the measured values and safe state indicators (e.g., fault flags, status and compare flags) from ADBMS6830 within the FTTI defined at system level (AoU00).

The ADBMS6830 does not take any action to put the system to safe state.

Table 1. Functional Safety Requirements

ID	Description
FSR01	Measurement inaccuracy of a cell voltage up to 4.5V must be detected at a magnitude no greater than ±15 mV. ASIL Target: D Fault Detection Time: <100 ms
FSR02	Measurement inaccuracy of a cell temperature up to 3V provided by external sensors must be detected at a magnitude no greater than ±15 mV. ASIL Target: D Fault Detection Time: <100 ms

SAFETY CONCEPT

The ADBMS6830 device primary functions are to measure in series connected battery cells voltages, to measure the voltage of the connected individual temperature sensors, and to communicate relevant measurement and status information to the BMS controller when it is requested. Multiple devices may be daisy-chained together using the isoSPI interface to create an efficient high-voltage battery monitoring system, or a single device may be connected to a suitable controller directly via a SPI bus. The safety mechanisms in this manual apply to both configurations. It is assumed that the item-level system is architected in terms of its hardware and software such that it can execute the ADBMS6830 safety requirements set forth in this manual. The ADBMS6830 functions may also be utilized by the system to help diagnose faults in the application circuit external to the IC.

ADBMS6830 combines safety mechanisms with redundancy in key functional blocks to efficiently achieve high ASIL metrics for both cell voltage measurements and temperature sensor measurements in addition to serial communication.
Key areas of redundancy

- Redundant measurement paths for cell voltage and GPIO measurements from input pins to the internal SPI controller
- Two high-precision references with diverse architectures (VREF1 and VREF2)
- Multiple independent oscillators,

Key safety mechanisms

- Open wire/path diagnostics on cells pins
- Internal power supply monitoring
- Communication CRC and frame transaction counter

The following section describes, which parts of ADBMS6830 fulfill the functionalities subject to functional safety requirements, derives their possible failure modes and develops a safety concept to achieve the functional safety requirement.

FSR01: CELL VOLTAGE MONITORING

Functional Description of Cell Voltage Measurements

The ADBMS6830 device features internal C-ADCs (C-ADCx in Figure 3), connected to the differential voltage inputs to measure individually cell voltages. Each C-ADC provides a 16 bit result every 1 ms which is subsequently corrected for ADC gain errors based on factory trim parameters. For further filtering, the ADC results are applied to subsequent CAVG8 (FIR) filters and to programmable IIR filters. All C-ADC results are stored in result registers accessible to the BMS controller via ADBMS6830's serial interface. The C-ADCs share a common reference and a common reference clock.

The cell voltage measurements can be impaired by the following Failure Mechanisms (FM):

FM1: Broken Input connection
FM2: Pad Leakage
FM3: Offset Error
FM4: Gain Error
FM5: Noise Error
FM6: Linearity Error
FM7: Gain Multiplication Error
FM8: ADC Filter Error
FM9: Storage Error
FM10: Execution Error
FM11: Communication Error
FM12: Oscillator Error
FM13: Supply Error
FM14: NVM/OTP Error
FM15: IIR Filter Errors

Functional Safety Architecture for Cell Measurement

Below is depicted the Safety Architecture for cell voltage measurements at the example of the x channel, where x could be from 1 to maximum supported cell channels of each device.

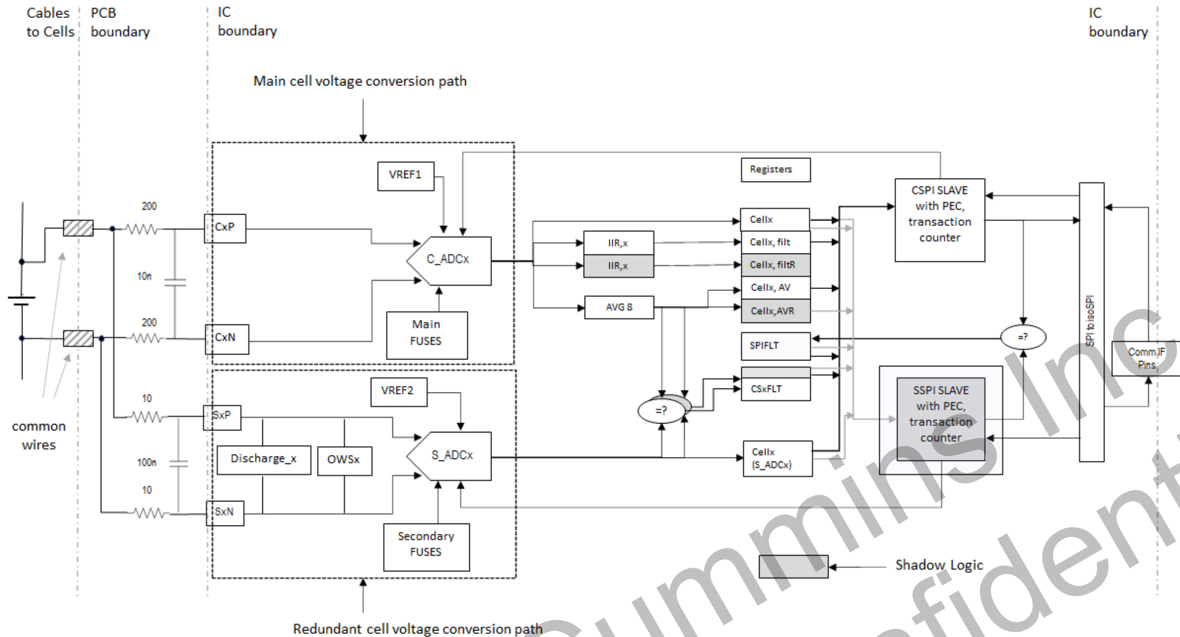

Figure 3. Cell Voltage Measurement Safety Architecture

ADBMS6830 device's functional safety concept regarding FSR01 is based on redundant measurement paths to measure all device supported cell voltages. Therefore, ADBMS6830 has as primary safety measure (SM_VCELL_RED) an additional set of S-ADCs (S-ADCx in Figure 3) with different architecture, allowing to measure the differential inputs at the SxP and SxN pins as shown in Figure 3. The Cx and Sx inputs are connect to the Cellx, where x values are from 1 to the maximum supported number of cells for each device.

The diversified S-ADCs have an update rate of 125 Hz and a resolution of 13 bits. They use a second reference and provide a redundant measurement path to measure cell voltages. When redundancy is enabled, the average of the last eight measurements of each C-ADC is compared with the 8 ms results of the respective S-ADC. If the result of a particular C-ADC and S-ADC channel couple differs by more than the CTH threshold, then the respective CSxFLT flag in the RDSTATC status register is set.

The AVG8 result registers are implemented redundantly. To ensure the integrity of the AVG8 result storage and transmission, the two AVG8 results are accessed and serialized by two separate SPI controller instances. The SM_SPI_RED compares the serialized MISO outputs. Mismatches between the two AVG8 results are flagged by the SPIFLT latch in the RDSTATC status register. Beside the averagers, the ADBMS6830 also provides IIR filters. The filter results are not compared with S-channel results. Rather, SM_IIR_RED implements the entire filter hardware redundantly. Computation, storage and transmission mismatches are reported through the same SPIFLT flag.

The result registers of the raw C-ADC and S-ADC cell voltage measurement results are not implemented redundantly. Data returned by the RDCV and RDSV commands is therefore not entirely safe, unless C-vs-S comparison is performed off chip.

Using these measurement paths introduces redundancy from the upfront external low pass filter down to the ADC result registers and therefore provides a safety mechanism allowing to detect FM1 (Broken Input), FM2 (Pad Leakage), FM3 (Offset Error), FM4 (Gain Error), FM5 (Noise Error), FM6 (Linearity Error), FM7 (Gain Correction Error), FM8 (ADC Filter Error), and also FM10 (Execution Error), as both measurement paths are controlled by separate State Machines, NVM bit array errors (FM14), IIR errors (FM15) and even communication errors (FM11) up to the SPI controllers. Furthermore, these SPI controllers add a pack error code (SM_SPI_PEC) (cyclic redundancy check CRC) to the data to be transmitted, allowing to detect downstream communication errors in the SPI to isoSPI conversion or in the connection to the BMS Controller. Communication errors leading to a non-detection of a command can be detected by reading the Command Counter (SM_SPI_CNT). Timely response of the device is verifiable by proper PEC and CNT values of the end of the received frames.

Oscillator Errors Supply Errors (FM13) are not covered by the redundant measurement paths. All the ADCs are clocked by the same oscillator.

To detect clock frequency errors, a second lower frequency oscillator is implemented and the numbers of clock cycles of the fast oscillator within a determined number of slow oscillator clock cycles are measured and reported (SM_CLOCK_MON). To detect a stuck clock a predefined sequence has to be executed (SM_FREEZE_SEQ).

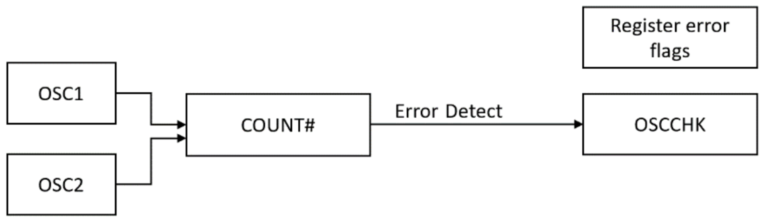

Figure 4. Clock Monitor

Supply Errors (FM13) are not covered by the redundant measurement paths, as the ADCs all use the same supply voltage.

To detect failures in the internal supply voltages, they are monitored and UV or OV are reported in the status register (SM_POWER_MON). In addition to the SM_POWER_MON, the 5V and 3V supplies are routed to AUXMUX and measured by the AUXADC (SM_POWER_OOR).

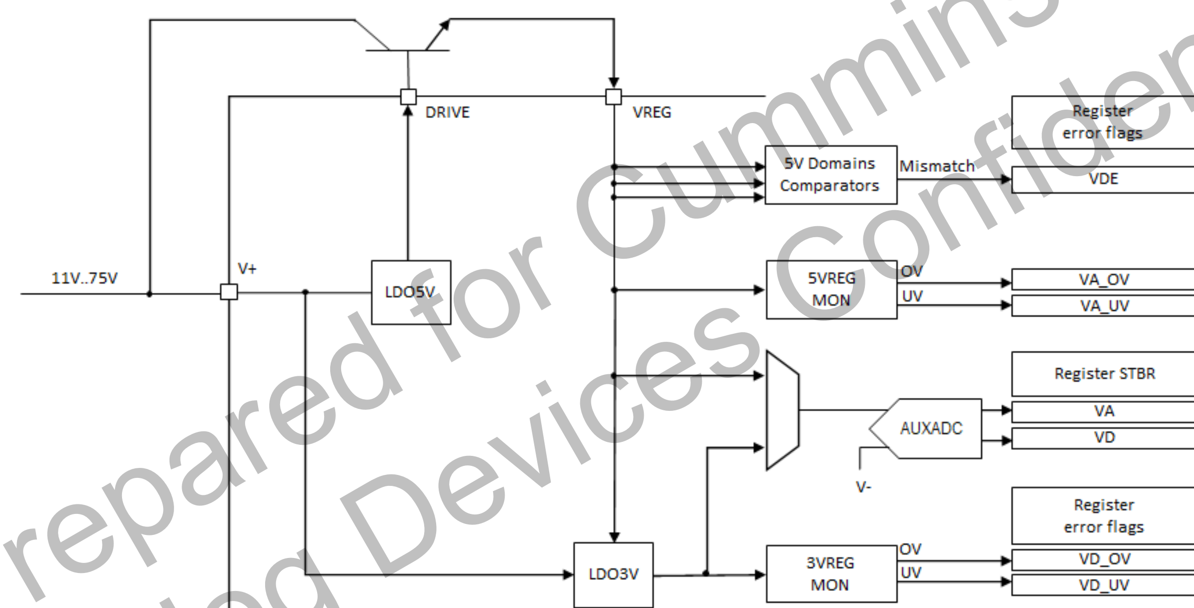

Figure 5. Power Supply Monitor

Storage errors (FM9) are largely but not fully covered by the redundant registers for every cell voltage result. Indeed, the update of all C-measurement path result registers are controlled by the same control signal for data coherency and signal data coherency failures will be detected by (SM_FREEZE_SEQ).

Erroneous activation of production Test Modes could contribute to several of the above-mentioned failure modes. Therefore, any activation of a test mode is indicated by a fault bit in the Status register (SM_TMOD_IND).

The sleep state is a low power consumption state with reduced functionality. Sleep mode entry will be flagged in a status register. The BMS controller can use the location to verify that during regular operation the IC has not erroneously entered the sleep state (SM_SLEEP_IND).

To prevent thermal damage, the ADBMS6830 devices feature a thermal shutdown feature, which automatically shuts down the device if the junction temperature is above the specified absolute maximum ratings, i.e., +150°C. The THSD flag reports the event. However, the operating range of the junction temperature is specified from -40°C to +125°C. Therefore, the safety concept does not rely on the shutdown feature as it is activated only when the temperature is already out of range. Rather, the die temperature is periodically checked using the AUXADC as described in SM_DIETEMP_MON.

Open Cell Voltage Sense Wires

Redundancy is used to detect on-chip failure modes of the cell measurement path according to SM_VCELL_RED. The redundant portions of the external application circuits for the cell inputs are diagnosed by the same concept. Applications often connect the individual battery cell poles with a single wire to the local PCB as shown in Figure 6, however meaning this part of the path must be checked by additional safety mechanisms. A short between wires is detectable by SM_VCELL_OOR. To reliably detect an open wire, the safety mechanism SM_CELL_OWD is needed.

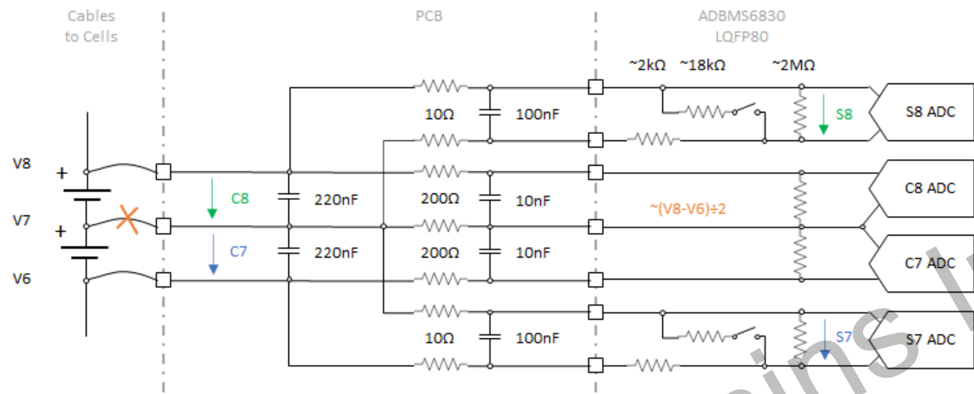

Figure 6. Open-Wire Detection on the Cell Voltage Sense Cables

During normal operation, the ADCs have high-impedance inputs in the 2 MΩ range. If the external wire was broken, the capacitors of the external filters would continue to store the last cell voltage before the wire broke for some time. Then, as shown in Figure 6, the input impedance of the ADCs presents a resistive divider that may pull the open circuit to a plausible-looking voltage, which is the average voltage of the neighbor cells. To ensure that these conditions are detected, the SM_CELL_OWD mechanism utilizes a switchable resistor divider between the input pins of the S-ADC channels. The total resistance is in the 20 kΩ range. If activated on only one of the channels connected to that cell voltage, it breaks the symmetry effect of the input impedances. Therefore, the open-wire switches are enabled on even and odd channels alternatingly. While the internal resistor divider is active, the S-ADCs measure a voltage that is attenuated by ~10%.

Figure 7 illustrates the signal behavior when the open-wire resistors are activated on the even channels for 8 ms first, and then for other 8 ms on the odd channels. It assumes that the cells C7 and C8 have the same initial voltage. In case of healthy connections, the battery cell will deliver the small current needed to support the resistive load. While the open-wire test is performed on the S-ADC channel, the C-channel can continue to measure the cell voltage without being disturbed, therefore preserving the integrity of the IIR filter result. The dotted lines in the left diagram prove that the external voltages C7 and C8 are not changing. For illustration the diagram intentionally shows a small delta that is not to scale. The bold lines show that during the first cycle the even channel S8 measures only 90% of the external voltage, and the odd channel S7 does so during the second cycle.

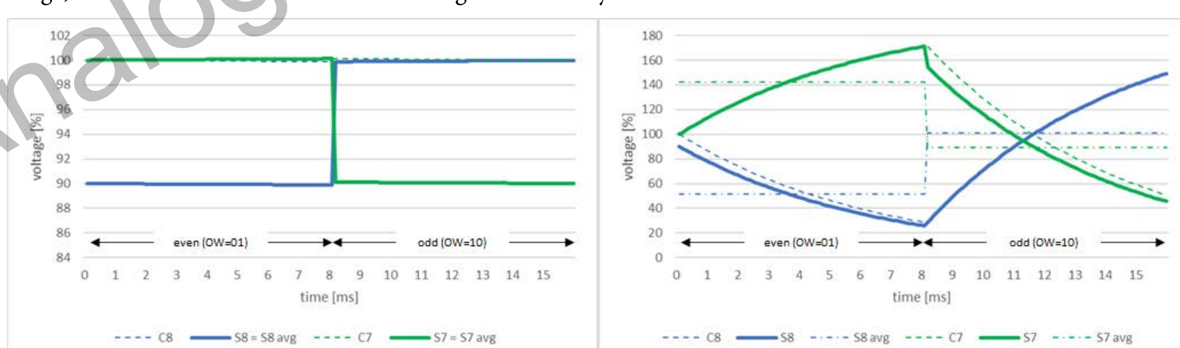

Figure 7. Open Wire Timing Diagrams for Tau = 320nF × 20 kΩ = 6.4 ms

In case of open wire, the open-wire resistors drain the capacitors as shown in the right-hand diagram of Figure 7. While the capacitor on the even channel drains, the voltage on the neighbor odd channel's capacitor raises by the same amount, because the delta between V8 and V6 does not change and the common wire between channel 7 and 8 is floating. When the 8 ms even cycle is over and the odd cycle starts, the S7 capacitor's voltage has increased vs. the original voltage. The odd cycle drains the C7 capacitors starting from that elevated point on as shown by the green dotted line. The C8 voltage responds with the inverted curve. The bold lines show the voltages presented to the S-ADCs.

The dash-dotted lines show the average value that the integrating ADCs are going to report after the 8 ms conversion time. One can see that a comparison of the even measurement results versus the original signal measurement result can detect an open wire. A 5-15% reduction suggests a good connection, a reduction that is down by more than 20-25% suggests an open wire. In case of significant system dynamics, it may be difficult to compare a measurement result with one that has been taken 8 ms earlier. Since a load change causes a common-mode response from two neighbor cells, the risk of false alarms can be minimized by further comparing the even response of S7 and S8 channels mutually.

The timing diagrams in Figure 7 are based on the capacitor values shown in Figure 6. Often, additional capacitance is seen on the lower cell channels. Where the cell voltage inputs are loaded with higher capacitor values, it takes longer to discharge the capacitors. The voltage drop caused by an 8 ms open-wire test, may become limited at one point. Then, detection margin can be gained, by performing two even or two odd open-wire cycles one after the other.

Whenever the S-channel is not operating in discharge mode for cell balancing, it can be commanded to alternate between SM_VCELL_RED and SM_CELL_OWD operation. The safety mechanism SM_CELL_OWD_DIAG confirms once per multi-point fault detection time interval that the diagnostic is indeed functional.

Open Supply Wires

The battery cells are not only connected through the cell voltage measurement sense lines. Typically, the ADBMS6830 power is supplied from the cells. Then, the top-most cell of the module is connected to the V+ input and to the NPN transistor that powers the VREG input. The minus pole of the lowest cell is connected to the local ground, supplying the V- and EPAD pins. To achieve full FSR and datasheet accuracy capability, voltage drop on the sensing wires must be avoided. Where the sensing wires do not conduct with negligibly low impedance, currents other than the ADC input current must not be allowed to flow through these wires. To prevent that the ADBMS6830 supply currents cause voltage drop on the cable, the V+ and V- supplies should be connected through separate wires. The options are limited for the REV3 QFN72 package, because it does not feature a dedicated S1N pin. If the wire impedance is not negligible, then the voltage drop must be compensated by application-level measures.

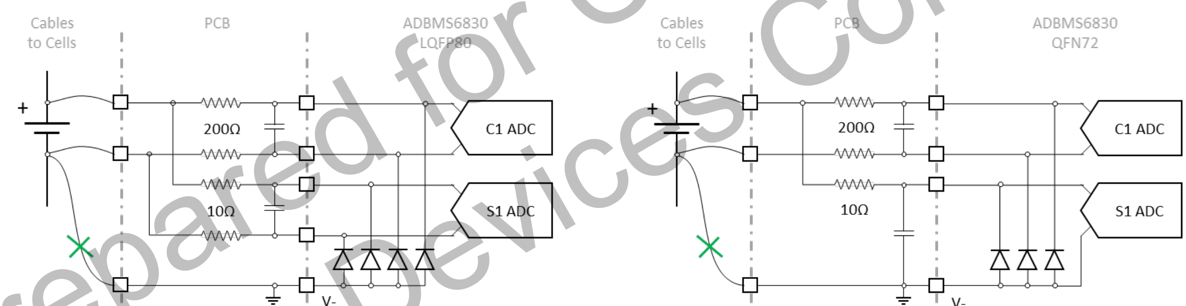

Figure 8. Open Wire Detection on the V- Supply Cable

For the V- connection one needs to distinguish between applications using the LQFP80 package and those using the QFN72 package. If the V- connection broke as illustrated in Figure 8, the device is no longer grounded properly. The supply current cannot escape through the V- wire and detour through the internal ESD protection diodes toward to sense lines instead. The internal device ground is lifted up until the diodes start conducting. While this is an immediate issue for the single-ended AUX ADC measurements, this is not yet causing the differential C-ADC and S-ADC operation to fail internally. However, the C1-ADC and the S1-ADC are seeing different voltages. In case of the QFN72 package this is because the voltage on the S1-ADC collapses. In case of the LQFP80 package this is because the internal diodes are not matched and because supply current causes different voltage drop on the external filter resistors. Therefore, in either package the SM_VCELL_RED safety mechanism will report this failure mode through the CS1FLT flag.

If the V+ connection breaks, then the device is no longer powered and any SPI communication including isoSPI daisy-chain relay function will fail. There are no hidden diodes on the device that would power V+ or VREG through any cell inputs. The safety concept relies on SM_SPI_PEC to report the resulting communication timeout in form of an CRC error. For additional availability one may install external diodes that power the device from alternative cell poles when the V+ wire breaks. Care is required in preventing such diodes from introducing undetectable common-cause failure modes because of capacitive noise coupling and because of voltage drops on the sense wires once the diode starts conducting. See AoU16 and AoU17.

High Common Mode ADC-Capacitors

The gain of the C- and the S-Measurement path both rely on metal-to-metal capacitors. The metal-to-metal capacitors are diversely implemented to avoid common cause faults on the main and redundant measurement path. The drift of these capacitors will be detected by the CAVG8-vs-S comparison (SM_VCELL_RED).

FSR02: TEMPERATURE MEASUREMENTS VIA GPIOS

Functional Description of Temperature Measurements via GPIOs

The ADBMS6830 device's general-purpose inputs/outputs (GPIOs) are connected through a multiplexer (AUX Multiplexer) to an auxiliary analog-to-digital converter, the AUX-ADC. Upon an ADAX trigger command, the AUX-ADC performs measurements through the inputs selected by the parameters of the ADAX command. When the ADC completes conversions, it updates the corresponding AUX result registers. It does not matter to the safety concept, whether the ADAX parameters command measurement of all channels in round-Robin fashion or whether channels are triggered individually, as long as each safety-relevant value is measured within the required diagnostic test interval.

The safety concept assumes that the temperature of the battery cells is measured through resistive temperature sensors, i.e., thermistors. Typically, the VREF2 output voltage is divided by a reference resistor and an NTC as shown in Figure 9. The voltage drop on the NTC is measured through the GPIO channels. The system integrator may extend the principle to other sensor types than considered by the ADBMS6830 safety concept.

The General-Purpose measurements can be impaired the following Failure Mechanisms:
FM1: Broken Input connection
FM2: Pad Leakage
FM3: Multiplexer Error
FM4: Offset Error
FM5: Gain Error
FM6: Noise Error
FM7: Linearity Error
FM8: Gain Multiplication Error
FM9: ADC Digital Filter Error
FM10: Storage Error
FM11: Execution Error
FM12: Communication Error
FM13: Oscillator Error
FM14: Supply Error
FM15: NVM/OTP Error

Functional Safety Architecture for GPIOs Measurements

The safety architecture for FSR02 is again based on redundancy. It is implemented by the safety mechanism SM_VGPIO_RED. First, it assumes that a pair of resistive dividers with thermistors (NTCs) is connected to two separate GPIO pins to measure the same temperature as shown in Figure 9. The nominal resistance of 10kΩ of the thermistors causes the measurement to be sensitive to leakage currents. A leakage current at the GPIO pad can only be detected by using a redundant external divider connected to a different GPIO input.
Internally of the ADBMS6830 device, a second measurement path allows to measure all GPIO inputs via the second multiplexer AUX2-MUX and the subsequent diverse analog-to-digital converter AUX2-ADC as shown in Figure 9.

By connecting two external redundant NTCs to two non-adjacent GPIO pins, and by using the internal redundant measurement paths, failure modes FM1 (Broken Input connection), FM2 (Pad Leakage), FM3 (Multiplexer Error), FM4 (Offset Error), FM5 (Gain Error), FM6 (Noise Error), FM7 (Linearity Error), FM8 (Gain Multiplication Error), FM9 (ADC Digital Filter Error), FM10 (Storage Error) and FM11 (Execution Error) can be observed by comparing the measurement results obtained via the two redundant measurement paths. The comparison is executed by the BMS controller.

Communication Error (FM12), Oscillator Error (FM13) and NVM/OTP Error (FM15) are detected by Safety Mechanisms described earlier in the context of FSR01. Also Supply Errors (FM14) are detected the same way. However, unlike the cell voltages that are measured differentially, the GPIO voltages are measured in single-ended fashion versus V-. This makes GPIO measurements more sensitive to unreliable ground connections. The SM_POWER_OOR mechanism is therefore enhanced to also confirm proper ground connection. SM_POWER_OOR_DIAG counterparts it to prevent multi-point faults.

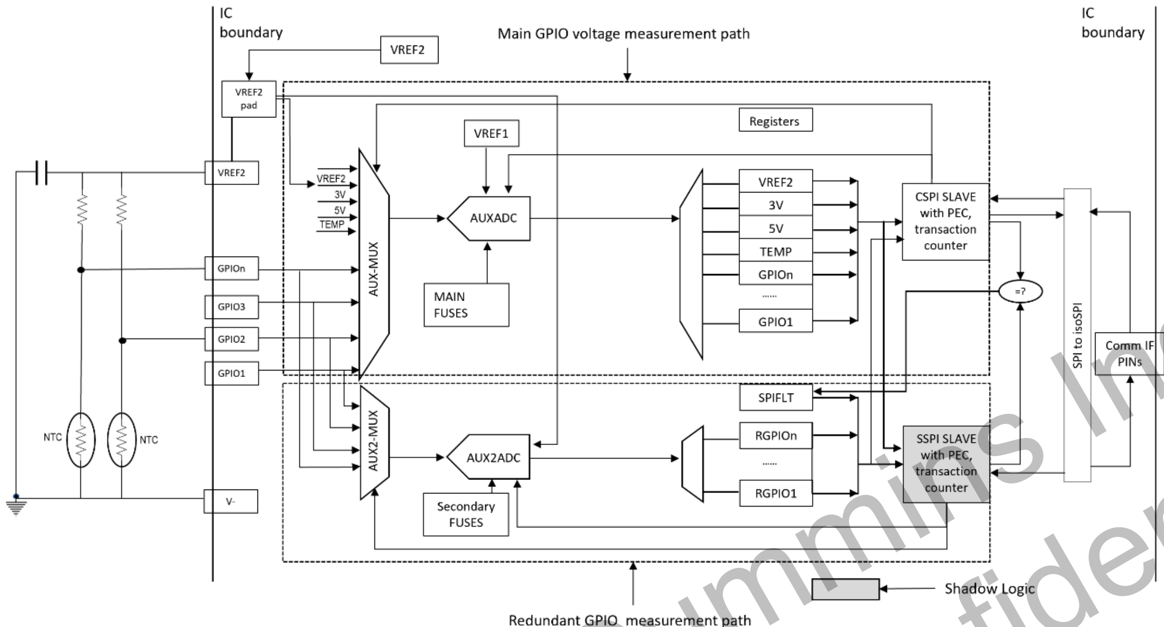

Figure 9. GPIO Measurement Safety Architecture

The internally generated VREF2 reference voltage is routed to the VREF2 pad. The output of the VREF2 pad is bonded to the VREF2 pin and also returned to the AUX2ADC and S-ADCs as reference voltages. If the VREF2 pad output is out of range, this fault is detected by SM_VCELL_RED. The failure modes of VREF2 pad and pin (including open or short to ground) are detected by the SM_VGPIO_OOR due to the incorrect VREF2 reference to the NTCs. The failure mode VREF2 pin leakage to VREG pin will result in incorrect VREF2 reference voltage to AUX2ADC and S-ADCs and this fault is detected by the SM_VCELL_RED and SM_VGPIO_RED.

The absolute value of the VREF2 output may vary from device to device slightly and also depends on the external components loading the pin. For temperature measurements the impact of the absolute VREF2 voltage can be ratiometrically minimized by dividing the NTC voltages measured through the GPIOs by the VREF2 measurement in software.

As an SEooC the ADBMS6830 safety concept assumes that two NTCs are connected externally to continue the redundancy concept off chip. Ideally, the NTCs are even thermally coupled to ease the comparison. In the context of a specific application the system integrator may install alternative plauzibilization strategies for the failure modes of the external NTCs.

If there were non-redundant NTCs remotely connected, the safety mechanism SM_VGPIO_OOR would detect shorted NTCs. It would also detect open wire, because of the high-side resistor, which is typically installed locally and would pull the input toward the VREF2 voltage. However, the on-chip failure modes were not fully protected. Where the system integrator decides to utilize a non-redundant NTC, that single NTC should be connected to two non-adjacent GPIO pins through separate series resistors. SM_VGPIO_RED is still required for the ADBMS6830 failure modes. The two series resistors ensure independence between the AUX and AUX2 channels and cause a voltage difference when open-wire detection currents are injected by the ADAX command.

Open Cell Temperature Sense Wires

The safety mechanism SM_VGPIO_OOR detects if any connection between the NTCs and the ADBMS6830 breaks. Using two NTCs connected to two separate, non-adjacent GPIOs measured by two separate ADCs makes open inputs fully detectable by SM_VGPIO_RED. The safety concept does not need to take advantage of the open-wire functionality of the AUX ADC for the purpose of truly redundant GPIO measurements.

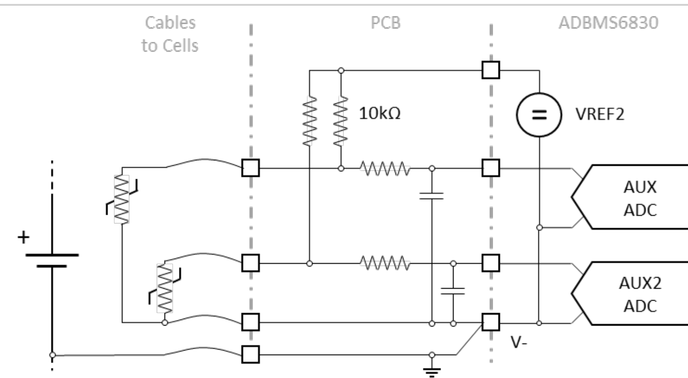

Figure 10. NTC Connectivity

The redundant SM_VGPIO_RED concept assumes individually connected NTCs. Where redundant NTC resistors share a common ground return wire as shown in Figure 10, any voltage drop on the common wire contributes to the common measurement error. There, the wire impedance needs to be controlled by systematic measures. The NTC resistors should not share the wire with the V- supply of the ADBMS6830. See AoU16. The safety mechanism SM_VGPIO_OOR detects if the common NTC ground return wire breaks. The SM_VCELL_RED mechanism detects broken V- connection. See Open Supply Wires.

Whereas the cell voltage is always measured differentially, the GPIO voltages including NTC voltages are measured in single-ended fashion versus device ground. Therefore, the GPIO measurement is more sensitive to unreliable V- and EPAD connections than the cell voltage measurement is. Where the HR pin is left floating on the PCB, unconnected ground pins will be detected by either SM_POWER_MON or SM_VCELL_RED. Whether the one or the other is the dominating mechanism depends on the package used. Where the HR pin is grounded on the PCB, the AUXADC based SM_POWER_OOR mechanism is mandatory. It requires any of the GPIO pins to be connected to ground. On the QFN package GPIO10 can be used for the purpose, as it is bonded to ground internally.

The safety mechanisms SM_POWER_OOR_DIAG uses the AUX ADC's open-wire feature for latent diagnostics, whether the one GPIO is indeed grounded as required by SM_POWER_OOR. There, the open-wire feature injects a ~200 µA test current into the AUX ADC that progresses to the one GPIO that is currently selected by the multiplexer. When the GPIO is grounded, the current causes a voltage drop on the ~550 µΩ output impedance of that pin. If the GPIO connection is open, then the test current cannot escape and the AUX ADC will measure a very high value.

FAULT FLAGS

The events listed in the below tables may not necessarily indicate malfunction of the chip. Many events may be caused by unexpected environmental conditions, unexpected external signals or by external circuit issues.

The default state of the fault flags is 1 (e.g., indicating a fault). Before starting the application, the BMS controller must clean them to a no-fault indicator state (e.g., clear to zero). The CLRFLAG command must be used to reset the fault flags in the Status Register Group C.

Table 2. Status Register C Flags

ID	Description
CSxFLT	Set by C vs S measurement detected mismatch Safety Mechanism: SM_VCELL_RED Flag: Status registers C, STCR0, STCR1 To Clear: Issue CRLFLAG command with CL_CSxFLT bit set. Latent Check: Activating Open-Wire switches during a redundant cell measurement.
VA_OV	Set by 5V voltage monitor in case of overvoltage on the analog supply. Safety Mechanism: SM_POWER_MON Flag: Status registers C, STCR4, Bit 7 To Clear: Issue CRLFLAG command with CL_VAOV bit set. Latent Check: SM_POWER_MON_DIAG

VA_UV	Set by 5V voltage monitor in case of undervoltage on the analog supply. Safety Mechanism: SM_POWER_MON Flag: Status register C, STCR4, Bit 6 To Clear: Issue CRLFLAG command with CL_VAUV bit set. Latent Check: SM_POWER_MON_DIAG
VD_UV	Set by 3V voltage monitor in case of undervoltage on the digital supply. Safety Mechanism: SM_POWER_MON Flag: Status register C, STCR4, Bit 4 To Clear: Issue CRLFLAG command with CL_VDUV bit set. Latent Check: SM_POWER_MON_DIAG
VD_OV	Set by 3V voltage monitor in case of overvoltage on the digital supply. Safety Mechanism: SM_POWER_MON Flag: Status register C, STCR4, Bit 5 To Clear: Issue CRLFLAG command with CL_VDOV bit set Latent Check: SM_POWER_MON_DIAG
VDE	Set when any of the 5V or GND domains outlies. Safety Mechanism: SM_POWER_MON Flag: Status register C, STCR5, Bit 6 To Clear: Issue CRLFLAG command with CL_VDE bit set. Latent Check: SM_POWER_MON_DIAG
VDEL	Set when all of the 5V or GND domains outlie. Safety Mechanism: n/a Flag: Status register C, STCR5, Bit 7 To Clear: Issue CRLFLAG command with CL_VDEL bit set. Latent Check: SM_POWER_MON_DIAG
CED	Single-bit Trim Error in C-NVM Safety Mechanism: Not applicable. Flag: Status register C, STCR4, Bit 3 To Clear: Issue CRLFLAG command with CL_CED bit set Note: the NVM ECC is required for reliability reason and not as a safety mechanism to protect the NVM bit array. The NVM bit array is implemented redundantly.
CMED	Multi-bit ECC Trim Errors in C-NVM Safety Mechanism: SM_NVM_ECC Flag: Status register C, STCR4, Bit 2 To Clear: Issue CRLFLAG command with CL_CMED bit set Latent Check: SM_NVM_ECC_DIAG
SED	Trim Error in S-NVM Safety Mechanism: Not applicable. Flag: Status register C, STCR4, Bit 1 To Clear: Issue CRLFLAG command with CL_SED bit set Note: The NVM ECC is required for reliability reason and not as a safety mechanism to protect the NVM bit array. The NVM bit array is implemented redundantly.
SMED	Multi-bit ECC Trim Errors in S-NVM Safety Mechanism: SM_NVM_ECC

	Flag: Status register C, STCR4, Bit 0 To Clear: Issue CRLFLAG command with CL_SMED bit set Latent Check: SM_NVM_ECC_DIAG
OSCCHK	Detects out-of-range oscillator count during ADC operation. Safety Mechanism: SM_CLOCK_MON Flag: Status register C, STCR5, Bit 0 To Clear: Issue CRLFLAG command with CL_OSCCHK bit set. Latent Check: SM_CLOCK_MON_DIAG
TMODCHK	Test Mode Indicator. Safety Mechanism: SM_TMOD_IND Flag: Status register C, STCR5, Bit 1 To Clear: Issue CRLFLAG command with CL_TMODE bit set Latent Check: SM_TMOD_IND_DIAG
THSD	Thermal Shutdown indicator. Safety Mechanism: n/a Flag: Status register C, STCR5, Bit 2 To Clear: Issue CRLFLAG command with CL_THSD bit set. Latent Check: SM_THSD_IND_DIAG
SLEEP	Sleep Indicator. Device has cycled through sleep mode. Safety Mechanism: SM_SLEEP_IND Flag: Status register C, STCR5, Bit 3 To Clear: Issue CRLFLAG command with CL_SLEEP bit set.
SPIFLT	Detected mismatch between redundant SPI controller outputs Safety Mechanism: SM_SPI_RED Flag: Status register C, STCR5, Bit 4 To Clear: Issue CRLFLAG command with CL_SPIFLT bit set. Latent Check: SM_SPI_RED_DIAG

SAFETY MECHANISMS AND DIAGNOSTICS

Table 3. Safety Mechanisms in Alphanumerical Order

ID	Name	ID	Name
SM_AUXREGS_DIAG	AUXADC Results Registers Latent Check	SM_SLEEP_IND	Sleep Mode Indicator
SM_CELL_OWD	Cell Open Wire Detection	SM_SPI_CNT	SPI Frame Counter
SM_CELL_OWD_DIAG	Cell Open Wire Detect Latent Check	SM_SPI_PEC	SPI Packet Error Code
SM_CFG_RED	Redundant Configuration Registers	SM_SPI_PEC_DIAG	PEC Algorithm Latent Check
SM_CLOCK_MON	Clock Monitor	SM_SPI_RED	Redundant SPI Controller
SM_CLOCK_MON_DIAG	Clock Monitor Latent Fault Check	SM_SPI_RED_DIAG	SPI Compare Logic and Fault Flag Indicator Check
SM_DIETEMP_OOR	Die Temperature Out-of-range Checker	SM_THSD_IND_DIAG	Thermal Shutdown Fault Flag Indicator Check
SM_FREEZE_SEQ	No Clock and Freeze Detector	SM_TMOD_IND	Test Mode Indicator
SM_IIR_RED	Redundant IIR Filters	SM_TMOD_IND_DIAG	Test Mode Fault Flag Indicator Check
SM_NVM_ECC	Single bit error correction, multi bit error detection on protected array	SM_VCELL_CMP_DIAG	C versus S Compare Fault Flag Indicator Check
SM_NVM_ECC_DIAG	ECC Latent Fault Check	SM_VCELL_OOR	Cell Voltage Out-of-range Checker
SM_POWER_MON	Supply Monitor	SM_VCELL_RED	Cell Voltage Measurement Redundancy
SM_POWER_MON_DIAG	Supplies Monitor Latent Check	SM_VGPIO_OOR	NTC Voltage Out-of-range Checker
SM_POWER_OOR	Voltage Supplies Out-of-range Checkers	SM_VGPIO_RED	Thermistor (GPIO) Voltage Measurement Redundancy
SM_POWER_OOR_DIAG	Supply Voltage ADC Latent Check	SM_VREF2_OOR	VREF2 Reference Voltage Out-of-range Checker

The description of the individual safety mechanisms is followed by a summary table. Therein the Target field has one of the options:

- Single point: The safety mechanism is targeting single-point faults;
- Multi point: The safety mechanism is targeting multi-point faults, including latent faults;
- Common cause: The safety mechanism is targeting common-cause faults identified by dependent failure analysis and may manifest themselves as single-point faults.
- Cascading: The safety mechanism is targeting cascading faults identified by dependent failure analysis and may manifest themselves as single-point faults.

The Type field indicates the nature of the mechanisms:

- Periodic: The safety mechanism is executed periodically, typically once per DTTI or FDTI, irrespective, whether the mechanism executes automatically, or has to be triggered from outside of the device;
- Continuous: The safety mechanism is either of online nature or is applied to all data when produced or consumed, irrespective, whether the mechanism executes automatically, or has to be verified outside of the device;
- On demand: The safety mechanism is executed in response to external trigger when needed, for example once per MPFDTI or every time when configuration has changed.

The Flags field indicates which latch bits are used for fault reporting.

The Variable field indicates the memory-mapped register, which provides results of the mechanism, for example the measured temperature value in case of the temperature check. Similarly, the Threshold field indicates the variables or registers defining the threshold for fault reporting.

Some safety mechanism require the system integrator to execute specific sequences, such as a series of SPI commands with respective check on returned data. It is recommended practice to:

- Whenever a sequence suggests to send a specific command for the purpose of setting a bit or bit field in a configuration register to a specific value, one should not alter the other bits belonging to the same configuration register. This can be ensured be either read-modify-write operations or by tracking the content of the configuration register by a software copy;
- Whenever a sequence suggests the CLRFLAG command, one should use the RDSTATC command to not miss any fault event pending before the sequence has even started;
- Unless it is for the purpose of testing STATC flags themselves, one should generally never blindly clear flags with the CLRFLAG command, without having inspected the flags with the RDSTATC command and without reporting events to the safe state controller, i.e., one should never clear a bit that was not set to a 1 by hardware.

Table 4. History of Safety Mechanisms

Rev PrE	Rev PrD	Rev PrC	Rev 0.2
SM_VCELL_RED	SM_VCELL_RED	SM_VCELL_RED	SM_RED_CELL_MEAS_PATH
SM_SPI_RED	SM_SPI_RED	SM_SPI_RED	SM_RED_SPI
SM_IIR_RED	SM_IIR_RED	SM_IIR_RED	SM_RED_IIR_FILTERS
SM_SPI_PEC	SM_SPI_PEC	SM_SPI_PEC	SM_PEC_SPI
SM_CLOCK_MON	SM_CLOCK_MON	SM_CLOCK_MON	SM_CLOCK_MON
SM_FREEZE_SEQ	SM_FREEZE_SEQ	SM_NOCLOCK_SEQ	---
SM_POWER_MON	SM_POWER_MON	SM_POWER_MON	SM_POWER_MON
SM_POWER_OOR	SM_POWER_OOR	SM_POWER_OOR	---
SM_CELL_OWD	SM_CELL_OWD	SM_CELL_OWD	SM_CELL_OWD
SM_VCELL_OOR	SM_VCELL_OOR	SM_VCELL_OOR	SM_VALID_RANGE_MEAS_VOLTAGE
SM_VGPIO_RED	SM_VGPIO_RED	SM_VGPIO_RED	SM_RED_TEMP_MEAS_PATH
SM_VGPIO_OOR	SM_VGPIO_OOR	SM_VGPIO_OOR	SM_VALID_RANGE_MEAS_TEMP
SM_SPI_CNT	SM_SPI_CNT	SM_SPI_CNT	SM_M2S_TRANS_COUNTER
SM_TMOD_IND	SM_TMOD_IND	SM_TMOD_IND	SM_TEST_MOD_ENTRY_IND
SM_DIETEMP_OOR	SM_DIETEMP_OOR	SM_DIETEMP_OOR	SM_DIE_TEMP_MON
SM_SLEEP_IND	SM_SLEEP_IND	SM_SLEEP_IND	SM_SLEEP_MOD_ENTRY_IND
SM_VREF2_OOR	SM_VREF2_OOR	SM_VREF2_OOR	SM_VREF2_to_ADC_CH
SM_SPI_PEC_DIAG	SM_SPI_PEC_DIAG	SM_SPI_PEC_DIAG	SM_DIAG_PEC
SM_POWER_MON_DIAG	SM_POWER_MON_DIAG	SM_POWER_MON_DIAG	SM_DIAG_POWER_MON
SM_CLOCK_MON_DIAG	SM_CLOCK_MON_DIAG	SM_CLOCK_MON_DIAG	SM_DIAG_CLOCK_MON
SM_SPI_RED_DIAG	SM_SPI_RED_DIAG	SM_SPI_RED_DIAG	SM_DIAG_SPI_COMP
SM_TMOD_IND_DIAG	SM_TMOD_IND_DIAG	SM_TMOD_IND_DIAG	SM_DIAG_TEST_MODE_FLAG
SM_THSD_IND_DIAG	SM_THSD_IND_DIAG	SM_THSD_IND_DIAG	SM_DIAG_TEMP_MON_FLAG
SM_VCELL_CMP_DIAG	SM_VCELL_CMP_DIAG	SM_VCELL_CMP_DIAG	SM_DIAG_STATUS&COMP_REG
SM_CELL_OWD_DIAG	SM_CELL_OWD_DIAG	SM_CELL_OWD_DIAG	SM_DIAG_CELL_OWS
SM_AUXREGS_DIAG	SM_AUXREGS_DIAG	---	---
SM_NVM_ECC	SM_NVM_ECC	---	---
SM_NVM_ECC_DIAG	SM_NVM_ECC_DIAG	---	---
SM_POWER_OOR_DIAG	---	---	---

Table 4 provides a history how the safety mechanisms have been added or renamed in the recent versions of this safety manual.

CELL VOLTAGE MEASUREMENT MEASURES

SM_VCELL_RED: Cell Voltage Measurement Redundancy

The safety mechanism SM_VCELL_RED provides redundancy to detect failures related to the cell voltage measurement. In conjunction with SM_IIR_RED and SM_SPI_RED it protects data accessed with the RDFCx and RDACx commands. The hardware is not designed to protect data returned by the RDCVx or RDSVx commands only. However, alternative safety concepts may compare RDCVx or RDSVx data outside of the device.

The ADBMS6830 device can measure the cell voltages with two redundant measurement paths, simultaneously. The primary C-ADC path and the secondary S-ADC path operate with diverse architecture. There are two methods to enable the cell voltage measurement on main and redundant path:

- In the direct method, the BMS controller sends an ADCV command with the RD parameter bit set to 1. This (re)starts C-ADC and S-ADC measurement on all channels. 8 ms later the CAVG8 has averaged eight measurement results from the C-ADCs. The S-ADCs have completed one conversion.
- The indirect method synchronizes the S-ADCs to the already running C-ADCs. The C-ADCs have already been started with the ADCV command with the CONT parameter bit set to 1. When receiving an ADSV command that also has the CONT parameter bit set to 1, the S-ADC measurements are delayed until the next CAVG8 boundary to align S-ADC measurements and CAVG8 averaging in time. This method does not disturb on-going continuous C-ADC operation any. Results are available after 8 ms to 16 ms after the reception of the ADSV command.

Either of the methods causes the hardware to automatically compare the CAVG8 and the S-ADC results for all channels, as soon as the S-ADC results are updated. The 16 CSxFLT flags in the Status Register Group C report whether the delta of the two results on any channel is greater than the threshold specified by the CTH[2:0] configuration bits. The COMP status bits confirms that the hardware comparison is active. Once set, the BMS controller must acknowledge the event by clearing the signaling CSxFLT flags with the CLRFLAG command.

The CTH[2:0] bit fields provide multiple threshold options. To avoid false alarms, the threshold must not be smaller than the sum of the absolute total measurement errors (TME) of the C-ADC and S-ADC channels. The sum of the chosen threshold plus the TME of the S-channel must not exceed the accuracy requirements of FSR01:

$$|TME_{CADC}| + |TME_{SADC}| \leq |THR_{CTH}| \leq |THR_{FSR01}| - |TME_{SADC}|$$

The TME of the C-ADC and S-ADC measurements depends on the expected voltage range of the battery cells and also on the device grading. See the ADBMS6830 datasheet for details.

Table 5. Cell Voltage Thresholds

V_{DIF} < 4.5V	TME_{CADC}	TME_{SADC}	THR_{CTH}
typ	±1 mV	±6 mV	9 mV
max	±2 mV	±7 mV	9 mV

Despite the worst-case TME_{SADC} specified in the datasheet, the 15 mV target of FSR01 can be statistically met up to a cell voltage range of 4.5V, if the threshold is set to 9 mV (CTH=010).

The diversification concept of the C-channel and S-channel measurements implies that also the external input filters are implemented in diversified fashion. While the C and S filters should match their Tau (R×C product) for similar impulse response, the S-channel filter resistor should be significantly smaller than the C-channel resistor to support the cell balancing scheme. Figure 3 shows the recommended values.

Table 6. SM_VCELL_RED Summary

System Integrator's Responsibilities	SM Facts
System integrator to limit cell voltage range to 4.5V for FSR01 target. PCB to not supply the device through cell voltage sense cables. PCB to diversify external C-channel and S-channel input filters. BMC to configure the cell voltage measurement with redundancy. BMC to set the CTH threshold to 010 for FSR01 target. BMC to check the CSxFLT flags. BMC to rely on data returned via RDFC or RDAC, but not on the one returned by RDCV or RDSV only.	Target: Single Point Type: Continuous Impacts: FSR01 Flags: CSxFLT Commands: RDSTATC Variables: STCR0,STCR1 Thresholds: CFGRA0.CTH Execution time: 8 ms to 16 ms

SM_VCELL_CMP_DIAG: C versus S Compare Fault Flag Indicator Check

The safety mechanism SM_VCELL_CMP_DIAG diagnose COMP flag for latent faults.

The CAVG8-vs-S compare logic is implemented by redundant hardware. The comparison status is forwarded to the redundant SPI controllers as in SM_SPI_RED. Failures in the compare logic and in the compare status flags are detected by the SPI channel comparators. Mismatches are reported through the SPIFLT fault flag.

The redundant CAVG8-vs-S comparison is only enabled, when the COMP status bit is set. Therefore, once per multi-point fault detection time interval, at any time convenient for the application, the COMP bit has to be checked using the RDSTATC command. The COMP bit must be confirmed to be indeed 1 when it is supposed to be 1, i.e., either

- after an ADCV command has been issued with the RD parameter set to 1; or
- after an ADSV command has been issued with the CONT parameter set to 1 while the CADC is operating in continuous mode.

Additionally, the COMP bit must be confirmed to become 0 again, after the S-ADC has stopped the redundant operation, i.e., either

- after an ADCV command has been issued with the RD parameter cleared to 0; or
- after an ADSV command has been issued with the CONT parameter cleared to 0; or
- after an ADSV command has been issued with the CONT parameter set to 1, while the C-ADC was not operating in continuous mode.

The hardware does not clear the COMP status bit just because of the completion of a single-shot, redundant ADCV operation.

While SM_VCELL_CMP_DIAG suffices for detection of dual-point faults, additional accumulation of undetected faults over time can be prevented with help of the SM_CELL_OWD_DIAG mechanism.

Table 7. SM_VCELL_CMP_DIAG Summary

System Integrator's Responsibilities	SM Facts
BMC to confirm COMP=0 after ADCV(RD=0). BMC to confirm COMP=1 after ADCV(RD=1).	Target: Multi point Type: On demand Impacts: FSR01 Flags: SPIFLT, COMP Commands: ADCV, RDSTATC Variables: n/a Thresholds: n/a

SM_VCELL_OOR: Cell Voltage Out-of-range Checker

The safety mechanism SM_VCELL_OOR detects common-cause failures in the cell battery voltage measurement not detected by the SM_VCELL_RED mechanism.

The system validates all ADC measurement results on cell voltage against range deemed plausible. The system must set thresholds for the min and max boundary of plausible cell voltage range depending on battery type, hardware configuration, use case, etc. Cell voltage measurements outside the valid ranges must be considered a fault condition and the BMS controller must put the system in safe state. This prevents also for reset and clear values of the result registers from being undetected. The safety concept sets the valid range from 0.1V and 4.5V. The system integrator is encouraged to further narrow that window according to application-specific characteristics.

Table 8. SM_VCELL_OOR Summary

System Integrator's Responsibilities	SM Facts
BMC to continuously apply range check to the measured cell voltages returned by either RDACx or RDFCx commands.	Target: Single point Type: Continuous Impacts: FSR01 Flags: n/a Commands: RDACx, RDFCx Variables: Thresholds: n/a Execution time: 0.5 ms

SM_IIR_RED: Redundant IIR Filters

The safety mechanism SM_IIR_RED provides redundancy to detect failures related to the Internal IIR filters used for further noise filtering of the C-ADCs output values.

The 16-bit results of the C-ADCs are fed to configurable IIR filters for further noise filtering. To ensure detection of faults in the IIR filters following the C-ADCs, the IIR filters are implemented redundantly and their results are read out by two separate SPI slaves whose outputs are compared. The IIR filter results are indirectly compared while reading the results from results IIR register to SPI controller interface. As illustrated in Figure 13 a mismatch between the main and the redundant filter will result in setting the SPIFLT bit during the readout process.

Table 9. SM_IIR_RED Summary

System Integrator's Responsibilities	SM Facts
BMC to check the SPIFLT flag after the Filtered Cell Voltage Registers A-F are read.	Target: Single Point Type: Continuous Impacts: FSR01 Flags: SPIFLT Commands: RDSTATC Variables: FCV(A-F)Rx, x = 0 to 5 Thresholds: n/a Execution time: 0.5 ms

SM_CELL_OWD: Cell Open Wire Detection

The safety mechanism SM_CELL_OWD provides a mechanism to detect open cell voltage sense connections between the battery cells and the PCB, where the cables are not implemented redundantly.

The principles of the concept are discussed in section Open Cell Voltage Sense Wires. The ADBMS6830 features open-wire switches (OWS) between the differential inputs of each S-ADC as shown in Figure 6. When the switches are closed, internal series resistors limit the current. The resistors also form voltage dividers to attenuate the voltages presented to the S-ADCs. During normal operation the switches are open and the S-ADCs measure the true input voltages. When the differential OWS switches are closed, however, the relevant pins are loaded, and the voltages presented to the S-ADCs are attenuated, typically by 10%, but in a range of 5% to 15%. In case of a broken sense wire, the OWS resistors discharge internal and external input capacitors, and any stored voltage will collapse quickly.

Figure 11 shows the timing of an algorithm which performs continuous uninterrupted cell measurements by means of the C-ADC and alternates the S-ADC between redundancy and open wire detection:

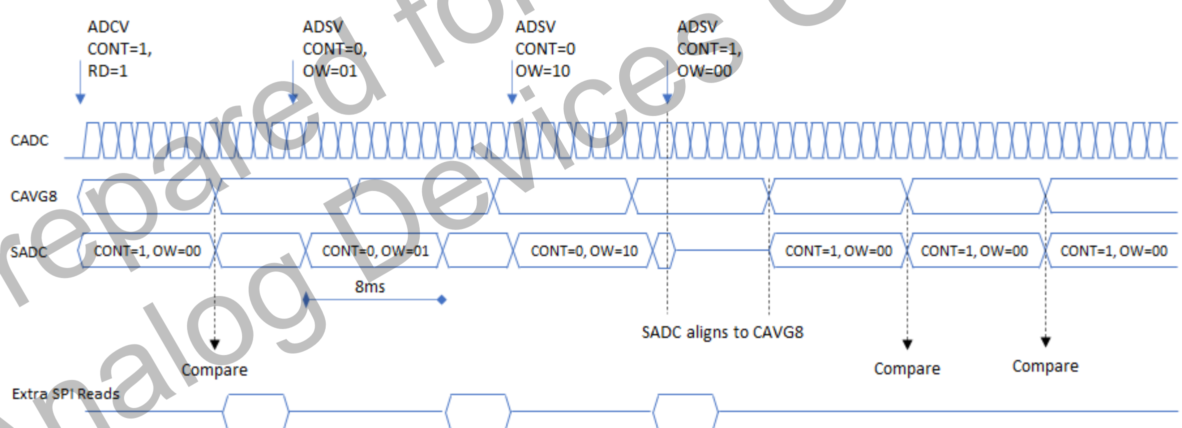

Figure 11. Cell Measurement and Diagnose Sequence

While the C-ADC measures the cell voltages continuously, the S-ADC is used sequentially to provide redundant measurement results and to perform open-wire detection periodically. Care must be taken, not to activate the differential switches of two neighboring channels simultaneously. Doing so would center the voltage on the open node between the voltages of the neighboring cells and the discharging of the capacitors would take longer. Therefore, the ADBMS6830 allows control of the open-wire switches of odd and even channels groups, so they can be activated in alternating fashion.

To perform an open-wire operation, send an ADSV command with either the OW parameter set as in Table 10, wait for 8 ms to let the operation complete and then send RDSVx to read the measurement results.

Table 10. Configure Open Wire Switches

OW[1:0]	Open Wire Switches
00	All channels off
01	Even channels on (S2, S4, …), Odd channels off (S1, S3, …)
10	Even channels off (S2, S4, …), Odd channels on (S1, S3, …)

OW[1:0]	Open Wire Switches
11	All channels on

The SM_CELL_OWD mechanism is performed by the following sequence:

1. Send ADSV with DCP=0, CONT=0, OW=00;
2. Wait for 8 ms;
3. Send RDSVx command to read all S-ADC results and store for reference;
4. Send ADSV with DCP=0, CONT=0, OW=01 (even open-wire check);
5. Wait for 8 ms;
6. Send RDSVx command to read all S-ADC results;
7. Send ADSV with DCP=0, CONT=0, OW=10 (odd open-wire check);
8. Wait for 8 ms;
9. Send RDSVx command to read all S-ADC results.

It is recommended to run the open-wire check (SM_CELL_OWD) right after the redundant measurement of SM_VCELL_RED. Then, to trigger the C-vs-S comparison, the first two steps in the above sequence are substituted by:

1. Send ADSV with DCP=0, CONT=1, OW=00 (redundant check);
2. Wait for 16 ms.

Where the cell voltage input pins are loaded with very large capacitors, the steps 4-6 and 7-8 may have to be repeated each to minimize risk of false alarms. For the capacitor values shown in Figure 6, the timing of the redundant and open-wire measurement will take 8-16 ms for the redundant measurement, 8 ms for the open-wire check on the odd channels and 8 ms for the open-wire check on the even channels. Thus, the whole redundant and open-wire diagnostic takes 24 ms to 32 ms. The read-out of the result registers adds some additional time depending on the isoSPI clock speed and daisy chain configuration.

As can be seen from Figure 7, the first check (the even) has the stronger signature than the subsequent one (the odd). The even check detects the open wire, irrespective whether it is on the plus or minus input of the channel. Since pin open failure modes are detected by SM_VCELL_RED, the odd check may be optional in some configurations. Still, it is recommended in the general case.

The results from the even channels gathered in steps 6 and 9 are compared against the ones the from step 3. A threshold set to ~75-80% of the nominal signal is reasonable and provides good immunity to false alarms due to system noise. Where applications may put more capacitance to the cell voltage inputs than shown in Figure 6, it is recommended to repeat the individual open-wire operations than setting the threshold even tighter. Generally, the optimal threshold value needs to be determined by analyzing all external components, wiring and signal dynamics. The lower the nominal cell voltage, the higher the relative contribution of the system noise. If the cell voltage becomes very small, then the relative attenuation of the open-wire measurements can hardly be judged. When the open-wire sequence is applied multiple times in a row, the voltage will collapse and settle to a low value. Therefore, the open-wire check should always be counterpartyed with an in-range check of the S-ADC results.

Table 11. SM_CELL_OWD Summary

System Integrator's Responsibilities	SM Facts
BMC to trigger open-wire detection for odd and even channels periodically. BMC to confirm relative attenuation of the open-wire results. BMC to confirm S-ADC results being in range.	Target: Single Point Type: Periodic Impacts: FSR01 Flags: n/a Commands: ADSV, RDSVx Variables: SV(A-F)Rx, CV(A-F)Rx, x = 0 to 5 Thresholds: see text

SM_CELL_OWD_DIAG: Cell Open Wire Detect Latent Check

The safety mechanism SM_CELL_OWD_DIAG diagnoses the open-wire switch circuits of SM_CELL_OWD for latent faults. It also confirms that the two redundant CAVG8-vs-S comparators are properly supplied with S-ADC results.

The principle of the SM_CELL_OWD_DIAG check is similar to the one for SM_CELL_OWD shown in Figure 11. The C-ADCs are enabled in the background and the S-ADCs take an open-wire measurement on odd and the even channels. While the SM_CELL_OWD is concerned whether the attenuation is more than 15%, the SM_CELL_OWD_DIAG is concerned about the minimal attenuation. If the S-ADC signal is attenuated by lesser than 5%, SM_CELL_OWD_DIAG fails and the SM_CELL_OWD cannot be trusted. As a latent fault mechanism, the SM_CELL_OWD_DIAG needs to execute only once per multi-point fault detection interval. It should run, when the battery system noise is low, such as during the startup, when the main contactors are still open. This way, the 5% threshold can be judged more accurately.

The 5% threshold assumes a settled signal. The open-wire switches are turned at the beginning of the conversion command. When selecting the threshold value for a specific application, the dynamic response of the external components to the ~20 kΩ load step needs to be considered in the context of the 8 ms integrating conversion time.

Unlike SM_CELL_OWD this latent fault check triggers the open-wire measurement with ADSV commands that have the CONT parameter set. These commands synchronize the S-ADC conversions with the CAVG8 of the C-ADC channels. At the cost of the synchronization latency, the open-wire S-ADCs results can be compared with the C-ADC results that are sampled at the same time. This makes the diagnostic further immune to dynamic changes of the input signal. Enabling the S-ADCs in continuous mode also has the side effect of additional benefit of the SM_VCELL_CMP_DIAG mechanism. The CONT parameter enables the CAVG8-vs-S channel comparison. The startup test can confirm that the CSxFLT flags of all odd and even channels are activated as expected. The CSxFLT flags need to be cleared with the CLRFLAG command when done.

The resulting algorithm looks like:

1. Send ADCV (RD=1, CONT=1, OW=00) command;
2. Send ADSV (CONT=1, OW=01) command and wait until conversion has completed (may take up to 16 ms);
3. Use RDACx commands to read AVG8 results;
4. Use RDSVx commands to read S-ADC results;
5. Confirm that the even results at least 5% down from the AVG8 ones and that the odd results are not;
6. Use RDSTATC command to confirm that all even CSxFLT flags are set, all odd CSxFLT flags are clear, COMP bit is set and SPIFLT is clear;
7. Send CLRFLAG command to clear all even CSxFLT flags;
8. Send ADSV (CONT=1, OW=10) command and wait until conversion has completed (may take up to 16 ms);
9. Use RDACx commands to read AVG8 results;
10. Use RDSVx commands to read S-ADC results;
11. Confirm that the odd results at least 5% down from the AVG8 ones and that the even results are not;
12. Use RDSTATC command to confirm that all odd CSxFLT flags are set, all even CSxFLT flags are clear, COMP bit is set and SPIFLT is clear;
13. Send CLRFLAG command to clear all odd CSxFLT flags;
14. Send ADSV (CONT=0) and wait until conversion has completed (may take up to 8 ms);
15. If wanted, send ADCV (CONT=0) to disable the C-ADCs again
16. Use RDSTATC command to confirm that COMP bit and SPIFLT are clear.

Table 12. SM_CELL_OWD_DIAG Summary

System Integrator's Responsibilities	SM Facts
BMC to execute above algorithm while contactors are open.	Target: Multi point Type: On demand Impacts: all FSR01 Commands: ADSV Flags: CSxFLT Variables: n/a Thresholds: n/a

GPIO MEASUREMENT MEASURES

SM_VGPIO_RED: Thermistor (GPIO) Voltage Measurement Redundancy

The safety mechanism SM_VGPIO_RED provides internal and external redundancy to detect failures related to the GPIO voltage measurement.

The battery cell temperatures are measured with the two independent and diverse AUX and AUX2 measurement channels, using redundant temperature sensors connected to separate GPIO pins as shown in Figure 9. See section Functional Safety Architecture for GPIOs Measurements for further background. The redundant temperature sensors can be connected to any of the GPIOs, as long as the following rules are followed:

1. The two redundant signals are connected to GPIOs that are not adjacent to each other;

2. At least one of the redundantly used GPIOs never activates the open-drain output in the given application;
3. At least one of the redundantly used GPIOs is not used for I2C or SPI communication in the given application;
4. Voltage drop on common ground return wires are controlled by the system integrator. See Figure 10.

It is up to the BMS controller to periodically trigger the conversions on AUX and AUX2 channels and to compare primary and secondary conversion result with each other. The ADBMS6830 safety concept is not aware on how closely external components are matched and thermally coupled, and can therefore not specify the optimal threshold for this comparison in a given application. As far as the ADBMS6830 itself is concerned, the threshold for the delta can be set to the sum of the total measurement errors (TME) of the two channels. Then, the total measurement error of the device's auxiliary channels can be optimized using the average of the two conversions weighted by the total measurement error of the individual channels:

- Delta = AUX_Result − AUX2_Result
- Threshold = AUX_TME + AUX2_TME
- Result = AUX_Result − Delta × AUX2_TME ÷ Threshold

The TME values for the AUX and AUX2 channels are specified in the datasheet. Since temperature measurements refer to the VREF2 voltage, the TME for the 3.3V input voltage range applies. The impact of the VREF2 accuracy to the NTC voltage has to be further considered and can be compensated with VREF2 reference measurements. See SM_VREF2_OOR.

Table 13. SM_VGPIO_RED Summary

System Integrator's Responsibilities	SM Facts
PCB to regard above GPIO selection rules for redundant temperature sensor connection. BMC to periodically trigger and read primary and secondary GPIOs voltages connected to the external thermistors. BMC to validate plausibility of primary AUX vs secondary AUX2 measurement results. System integrator to control impedance of common ground return wires.	Target: Single Point Type: Continuous Impacts: FSR02 Flags: n/a Commands: ADAX, ADAX2, RDAUXx, RDRAXx Variables:GPxRy,RGPxRy, x=A-D, y=0-5 Thresholds: n/a Execution time: 8.5 ms per GPIO pair

SM_VGPIO_OOR: NTC Voltage Out-of-range Checker

The safety mechanism SM_VGPIO_OOR detects common-cause failures in the cell temperature (GPIOs) voltage measurement not detected by the SM_VGPIO_RED mechanism.

The voltage drop on the NTC resistors can significantly vary with temperature, but will never be as high as VREF2 and never be as low as 0.1V. The safety mechanism checks in software whether measured voltage suggests a temperature within the expected range. A short of the NTC resistor will manifest itself as a voltage which is almost zero. Similarly, a short of the pull-up resistor will drive the voltage very close to the VREF2 value. If the NTC is open, the measured voltage is dominated by the resistor divider between the pull-up and the input resistance. If the pull-up is open the input drives to 0V.

The thresholds depend on the application and to a minor degree also on the external load of the VREF2 pin, and may therefore be a function of the VREF2 measurement gained by SM_VREF2_OOR.

Table 14. SM_VGPIO_OOR Summary

System Integrator's Responsibilities	SM Facts
BMC to continuously apply range check to the measured NTC voltages.	Target: Single point Type: Continuous Impacts: FSR02 Flags: n/a Commands: RDAUX(A-D) Variables: GP(A-D)Rx, x = 0 to 5 Thresholds: n/a Execution time: 0.5 ms

DEVICE MONITORING

SM_POWER_MON: Supply Monitor

The safety mechanism SM_POWER_MON detects if the 5V VREG supply voltage or the internal 3V supply voltage are out of range during ADC conversions. The SM_POWER_MON comparators are fast and can detect spikes. For accurate under- and overvoltage monitoring they partner with the SM_POWER_OOR mechanism, that periodically measures the 3V and 5V supply voltages with the accurate AUX ADC.

There are two main linear regulators, LDO5V and LDO3V, that provide supply for internal blocks in the ADBMS6830. The LDO5V supplies the DRIVE pin, which is needed to generate the VREG input voltage. The undervoltage and overvoltage conditions in the 5V VREG and 3V internal supply rails are monitored by independent analog window comparators. While ADC conversions are ongoing, the comparator outputs update the VA_OV, VA_UV, VD_OV and VD_UV flags in the Status Register Group C.

- The LDO3V monitor reports undervoltage below 2.6V ± 0.2V and overvoltage above 3.8V ± 0.2V.
- The LDO5V monitor reports undervoltage below 4.4V ± 0.1V and overvoltage above 5.65V ± 0.15V.

Internally, the input 5V VREG voltage is split into three bonding domains. Special delta monitors compare the three domain voltages mutually and report any mismatch through the VDE flag. Similarly, the input 0V V- voltage is split in four bonding domains. Again, mismatches are reported through the same VDE flag.

If any on the VA_OV, VA_UV, VD_OV, VD_UV, VDE bits is set to 1, the conversion results have to be considered being unsafe. Once set, the BMS controller needs to clear them with the CLRFLAG command to continue power monitoring. As long as the fault condition is still pending and any ADC conversion is still ongoing, hardware prevents the flags from being cleared. When all ADC conversions have finished, the bits can be cleared even if the fault condition is still pending.

Table 15. SM_POWER_MON Summary

System Integrator's Responsibilities	SM Facts
BMC to read each time after an ADC conversion the Flags and check that the Flags are not set indicating a fault.	Target: Single Point Type: Continuous Impacts: all FSRs Flags: VA_OV, VA_UV, VDE, VD_OV, VD_UV Commands: RDSTATC Variables: n/a Thresholds: n/a Execution time: 0.5 ms

SM_POWER_MON_DIAG: Supplies Monitor Latent Check

The safety mechanism SM_POWER_MON_DIAG diagnose SM_POWER_MON for latent faults.

The Internal Supply Fault Monitoring can be verified by a diagnostic test, proving diagnostic coverage for latent fault. During the test, an offset is added to the input signal of the comparators to force trigger.

The BMS controller must take following steps:

1. Send the CLRFLAG command to clear any pending fault flag bit, if any;
2. Send the WRCFGA command to set the FLAG_D[3:2] configuration bits to 01 for UV check;
3. Send a single-shot ADCV or ADAX command and wait for the conversion to complete;
4. Send the RDSTATC command to confirm that VA_UV and VD_UV flag bits are both set and all other fault flags are clear;
5. Send the WRCFGA command to clear the FLAG_D[3:2] configuration bits to 00;
6. Send the CLRFLAG command to clear the VA_UV and VD_UV flag bits;
7. Send the WRCFGA command to set the FLAG_D[3:2] configuration bits to 11 for OV and delta check;
8. Send a single-shot ADCV or ADAX command and wait for the conversion to complete;
9. Send the RDSTATC command to confirm that VA_OV, VD_OV, VDE and VDEL bits are all set and all other fault flags are clear;
10. Send the WRCFGA command to clear the FLAG_D[3:2] configuration bits to 00;
11. Send the CLRFLAG command to clear the VA_OV, VD_OV, VDE and VDEL flag bits;
12. Send the RDSTATC command to confirm that all fault flags are clear.

Table 16. SM_DIAG_ POWER_MON Summary

System Integrator's Responsibilities	SM Facts
BMC to execute the above sequence.	Target: Multi point Type: On demand Impacts: all FSRs Flags: VDEL, VD_UV. VD_OV, VA_UV. VA_OV Commands: CLRFLAG, WRCFGA, ADCV or ADAX, RDSTATC Variables: n/a Thresholds: n/a

SM_POWER_OOR: Voltage Supplies Out-of-range Checkers

In contrast to the SM_POWER_MON, which is fast and continuous, the safety mechanism SM_POWER_OOR accurately detects if the 5V VREG supply voltage or the internal 3V supply voltage is out of range by periodic sampling. The SM_POWER_OOR also detects faulty ground connections which could impact the single-ended GPIO voltage measurements and could lead to common-cause failures harming SM_VGPIO_RED.

The VREG input and LDO3V output are routed to the SM_POWER_MON hardware monitors and also to the AUX-MUX, so that the AUXADC can measure their voltage when requested by the BMS controller. The ADAX command with the CH[4:0]=b10001 parameter triggers an LDO3V measurement, and measures VREG when called with the CH[4:0]=b10010 parameter. Alternatively, the CH[4:0]=b00000 parameter can be used to trigger all measurements in a row. Once the conversions have completed, the RDSTATB command returns measurement results following the formulas:

Digital Supply (LDO3V output) = VD[15:0] × 150 μV + 1.5V

Analog Supply (LDO5V output = VREG input) = VA[15:0] × 150 μV + 1.5V

Where the HR pin is connected to V- on the PCB, the SM_POWER_OOR mechanism also needs to monitor also the voltage of a grounded GPIO pin. This is not required where the HR pin is left floating.

On the QFN package the GPIO10 input is bonded to V- internally of the package. In case of the LQFP package with the HR tied to V-, also the GPIO10 input, or any other spare GPIO, needs to be connected to V- on the PCB, either directly or through the ground plane. An AUX ADC measurement of that GPIO voltage confirms that the device is properly grounded. A redundant AUX2 ADC measurement of the same GPIO voltage ensures that latent faults in the AUX measurement are not left undetected. The GPIO voltage measurements are triggered with the ADAX and ADAX2 commands. In case of GPIO10 being the chosen GPIO, the results are available in the G10V and R_G10V variables. The voltage is determined by the formula:

AUX GPIO10 voltage = G10V[15:0] × 150 μV + 1.5V

AUX2 GPIO10 voltage = R_G10V[15:0] × 150 μV + 1.5V

Table 17. Supply Voltage Thresholds

Supply Rail	Supply Voltage MIN	Supply Voltage MAX	AUXADC Threshold MIN	AUXADC Threshold MAX
LDO5V	4.5V	5.5V	+4512 mV	+5486 mV
LDO3V	2.7V	3.6V	+2754 mV	+3528 mV
GPIO10	0V	0V	-5 mV	+5 mV

The accuracy of the AUXADC is specified as ±0.25% for the 5V supply measurement and as ±2% for the 3V supply measurement. Therefore, the thresholds are to be set as shown in Table 17.

Table 18. SM_POWER_OOR Summary

System Integrator's Responsibilities	SM Facts
BMC to periodically trigger the 3V and 5V measurements, read the results and check whether the measured values are in the ranges defined in Table 17 Where HR is connected to V-: PCB to connect GPIO10 to V- on the PCB (LQFP only); BMC to periodically trigger GPIO10 measurement, read the AUX and AUX2 results and check whether in range; BMC to keep the GPO10 open-drain control bit set.	Target: Single point Type: Periodic Impacts: all FSRs Flags: n/a Commands: ADAX, ADAX2, RDSTATB, RDAUXD, RDRAXD, WRCFGA Variables: VA, VD, G10V, R_G10V Thresholds: see Table 17 Execution Time: 2.5 ms

SM_POWER_OOR_DIAG: Supply Voltage ADC Latent Check

The safety mechanism SM_POWER_OOR_DIAG detects if the GPIO pin used in SM_POWER_OOR is not properly connected to the V-.

This safety mechanism uses the open-wire function of the AUXADC to inject a test current into the GPIO10 pin, or alternatively chosen GPIO pin. When the ADAX command is sent with both the OW and PUP parameter bit set, a test current is injected into the one GPIO pin that is currently selected by the AUX multiplexer. When the GPIO is properly connected to ground, the test current causes a small voltage drop on the internal resistance of the pin.

- If the measured voltage is between 50 mV and 500 mV the GPIO connection is proper;
- If the measured voltage is above 500 mV, the GPIO connection is not reliable;
- If the measured voltage is below 50 mV, the open-wire function is not reliable.

The open-drain transistor of the respective GPIO must be disabled during the test.

1. Send the WRCFGA command to set the GPO[10] bit;
2. Send the ADAX command with both the OW and PUP parameter bits set and CH set to 0x00 or to 0x0A;
3. Wait for the AUXADC conversion(s) to complete;
4. Send the RDAUXD command to confirm that the G10V value is between 50 mV and 500 mV.

Table 19. SM_POWER_OOR_DIAG Summary

System Integrator's Responsibilities	SM Facts
PCB to connect GPIO10 to V- on the PCB (LQFP only). BMC to enable test current injection to GPIO10 once at startup and verify whether the AUX results are in range. BMC to keep the GPO10 open-drain control bit set.	Target: Multi point Type: On demand Impacts: FSR02 Flags: n/a Commands: ADAX, WRCFGA, RDAUXD Variables: G10V Thresholds: 50 mV, 500 mV

SM_VREF2_OOR: VREF2 Reference Voltage Out-of-range Checker

The safety mechanism SM_VREF2_OOR detects failures related to the buffered VREF2 reference voltage. Since the measurement is taken by the AUXADC using VREF1 as a reference, it also detects VREF1 faults.

The ADBMS6830 has two independent diverse generated voltage references, VREF1 and VREF2. The primary reference, VREF1, provides a reference for C-ADCs and AUXADC analog converter. The VREF2 is a secondary reference for S-ADCs and AUX2ADC converters, as well for external thermistor circuits. The VREF1 is based on buried-junction Zener diode, providing superior stability over mechanical, temperature, and environmental variations. VREF2 is a standard bandgap reference with low drift over temperature.

The VREF2 reference voltage is connected to two inputs of the AUXADC multiplexer internally: It is connected directly to the VREF2 input and connected through a series resistor to the VRES input. The VRES input could be used to detect when the AUXADC's open-wire test current is enabled by a voltage drop observed relative to VREF2. For this safety mechanism, however, the VRES path is used because the reset and clear values of the VRES result register give better multi-point diagnostic coverage than the VREF2 register as described in SM_AUXREGS_DIAG. Because the open-wire test currents are not enabled during this test, the VRES measurement and VREF2 registers will give the same measurement results:

$$\text{VREF2 Reference Voltage} = \text{VREF2}[15:0] \times 150\ \text{mV} + 1.5\text{V} = \text{VRES}[15:0] \times 150\ \text{mV} + 1.5\text{V}$$

Unlike the VREF2 result register, the VRES result register responds with different values to reset and clear (CLRAUX) operation and SM_AUXREGS_DIAG applies. Therefore, the VRES result register has the better multi-point diagnostic coverage and the SM_VREF2_OOR safety mechanism relies on the VRES channel rather than on the VREF2 channel.

Table 20. VREF2 / VRES Monitoring Thresholds

Signal	Voltage MIN	Voltage MAX	AUXADC Threshold MIN	AUXADC Threshold MAX
VREF2 / VRES	2.994V	3.006V	2.985V	3.015V

The thresholds for the out-of-range check are defined in Table 20, considering the nominal VREF2 range and the AUXADC measurement accuracy.

A VRES measurement is triggered by the ADAX command with either parameter CH[4:0]=b00000 or CH[4:0]=b10110. Once the conversion has completed, the RDSTATB read command returns the VRES conversion result.

Table 21. SM_VREF2_OOR Summary

System Integrator's Responsibilities	SM Facts
BMC to trigger the VRES measurement, read the results and check if the measured value is in the ranges defined in Table 20.	Target: Multi Point Type: On demand Impacts: all FSRs Flags: n/a Commands:ADAX, RDSTATB Variables: VRES Thresholds: n/a Execution Time: 1.5 ms

SM_DIETEMP_OOR: Die Temperature Out-of-range Checker

The safety mechanism SM_DIETEMP_OOR detects when the junction temperature of the device gets out of the operational range.

The safety mechanism SM_DIETEMP_OOR detects when the junction temperature of the device gets out of the operational range. The ADAX command along with the channel configuration CH[4:0]=b10011 triggers the measurement of the die temperature sensor output, TEMP, scaled to a voltage output and connected to the AUXADC converter. Once the conversion is completed, the RDSTATA command reads the result from STAR2 and STRA3 registers, bits ITMP[15:0].

The formula to calculate the real die junction temperature from the ITMP result register:

- Temperature Measurement Voltage = (ITMP × 150 mV + 1.5V) /7.5 mV/°C – 273°C

Table 22. SM_DIETEMP_OOR Summary

System Integrator's Responsibilities	SM Facts
BMC to periodically trigger Die Temperature measurement and to evaluate measurement results for being in operational range, defined in the device data sheet.	Target: Common cause Type: Periodic Impacts: all FSRs Flags: n/a Commands: ADAX, RDSTATA Variables: ITMP Thresholds: n/a Execution time: 1.5 ms

SM_AUXREGS_DIAG: AUXADC Results Registers Latent Check

The safety mechanism SM_AUXREGS_DIAG diagnose the AUXADC results registers for latent faults.

After power-on reset or an SRST command alternatively, the AUXADC result registers VA, VD, VRES and ITMP reset to 0x7FFF value. Before issuing the first ADAX command, the BMS controller reads these values with the RDSTATA and the RDSTATB and confirms the reset value being correct. Then, the BMS controllers sends an CLRAUX command, which toggles all register bits to the 1's-complement value 0x8000. Again, the BMS controller confirms the 0x8000 values. Subsequent AUXADC conversions of constant results as in SM_DIETEMP_OOR, SM_POWER_OOR and SM_VREF2_OOR confirm that the result registers are indeed updating as expected.

1. Send the RDSTATA command to confirm ITMP = 0x7FFF;
2. Send the RDSTATB command to confirm VD = VA = VRES = 0x7FFF;
3. Send the CLRAUX command;
4. Send the RDSTATA command to confirm ITMP = 0x8000;
5. Send the RDSTATB command to confirm VD = VA = VRES = 0x8000;

The CLRAUX command requires the OSC1 clock to be up. Therefore, ensure that a WRCFGA command with the REFON bit set was sent no later than t_{REFUP} prior to the sequence.

Table 23. SM_AUXREGS_DIAG Summary

System Integrator's Responsibilities	SM Facts
BMC to confirm 0x7FFF values after reset. BMC to confirm 0x8000 values after CLRAUX.	Target: Multi point Type: On demand

System Integrator's Responsibilities	SM Facts
BMC to execute SM_POWER_OOR, SM_VREF2_OOR and SM_DIETEMP_OOR afterward.	Impacts: all FSRs Flags: n/a Commands: CLRAUX, RDSTATA, RDSTATB Variables: VA, VD, ITMP, VRES Thresholds: n/a

SM_CLOCK_MON: Clock Monitor

The safety mechanism SM_CLOCK_MON detects frequency errors in the main clock oscillator.

During any ADC conversions, the 2 MHz main oscillator is monitored by an independent 8 kHz oscillator. The number of the ADC clock cycles are counted during every half period of the 8 kHz clock. The monitor sets the OSCCHK flag (see Figure 4), if the clock count does not fall within ±15% of the target value.

If set to 1, the OSCCHK flag must be cleared by the CLRFLAG command for subsequent clock fault monitoring.

Table 24. SM_CLOCK_COMP Summary

System Integrator's Responsibilities	SM Facts
BMC to periodically inspect the status of the OSCCHK flag.	Target: Single point Type: Periodic Impacts: all FSRs Flags: OSCCHK Commands: RDSTATC Variables: n/a Thresholds: n/a Execution time: 0.5 ms

SM_CLOCK_MON_DIAG: Clock Monitor Latent Fault Check

The safety mechanism SM_CLOCK_MON_DIAG diagnose the SM_CLOCK_MON for latent faults.

The Internal Clock Monitoring can be verified by a diagnostic test, proving diagnostic coverage for latent faults by means of forcing the oscillator counter to be faster or slower.

The BMS controller must take following steps:

1. Send the CLRFLAG command to clear any pending fault flag bit, if any;
2. Send the WRCFGA command to set the FLAG_D[1:0] configuration bits to 01 to force the oscillator counter to run faster;
3. Send any ADC command to update the OSCCHK status registers;
4. Send the RDSTATC command to confirm that OSCCHK is set to 1;
5. Send the RDSTATD command to confirm that OC_CNTR[7:0] is above 70;
6. Send the CLRFLAG command to clear the OSCCHK flag;
7. Send the WRCFGA command to set the FLAG_D[1:0] configuration bits to 10 to force the oscillator counter to run slower;
8. Send any ADC command to update the OSCCHK status registers;
9. Send the RDSTATC command to confirm that OSCCHK is set to 1;
10. Send the RDSTATD command to confirm that OC_CNTR[7:0] is below 52;
11. Send the WRCFGA command to set the FLAG_D[1:0] configuration bits to 00 for normal operation;
12. Send the CLRFLAG command to clear the OSCCHK flag;
13. Send the RDSTATC command to confirm that OSCCHK is cleared;
14. Send any ADC command to update the OC_CNTR[7:0];
15. Send the RDSTATD command to confirm that OC_CNTR[7:0] is in between 52 and 70.

Table 25. SM_CLOCK_MON_DIAG Summary

System Integrator's Responsibilities	SM Facts
BMC to execute the above sequence.	Target: Multi point Type: On demand Impacts: all FSRs Flags: n/a

System Integrator's Responsibilities	SM Facts
	Commands: CLRFLAG, WRCFGA, ADCV, RDSTATC, RDSTATD Variables: n/a Thresholds: n/a

SM_THSD_IND_DIAG: Thermal Shutdown Fault Flag Indicator Check

The safety mechanism SM_THSD_IND_DIAG diagnose THSD flag for latent faults.

To confirm that the THSD diagnostic bit is not stuck, the BMS controller takes the following steps:

1. Send the CLRFLAG command to clear any pending fault flag bit, if any;
2. Send the WRCFGA command to set the FLAG_D[4] configuration bit to force the THSD bit;
3. Send the RDSTATC command to confirm that the THSD flag is set;
4. Send the WRCFGA command to clear the FLAG_D[4] configuration bit again for normal operation;
5. Send the CLRFLAG command to clear the FLAG_D[4] for normal operation;
6. Send the RDSTATC command to confirm that the THSD flag is clear.

Table 26. SM_THSD_IND_DIAG Summary

System Integrator's Responsibilities	SM Facts
BMC to execute the above sequence.	Target: Multi point Type: On demand Impacts: all FSRs Flags: n/a Commands: CLRFLAG, WRCFGA, RDSTATC Variables: n/a Thresholds: n/a

SM_TMOD_IND: Test Mode Indicator

The safety mechanism SM_TMOD_IND indicates unintended activation of test modes.

The ADBMS6830 features test modes for the solely purpose of factory testing. They are activated by keyed sequences of hardware and software steps. The TMODCHK flag indicates that any of the test modes is active, and warns also in case of any being activated inadvertently. The bit is always set after reset and must be initially cleared by the CLRFLAG command. If it cannot be cleared or if set afterward again, the device must be considered being unsafe.

Table 27. SM_TMOD_IND Summary

System Integrator's Responsibilities	SM Facts
BMC to check if the TMODCHK is set after an ADC conversion.	Target: Common cause Type: Periodic Impacts: all FSRs Flags: TMODCHK Commands: RDSTATC Variables: n/a Thresholds: n/a Execution time: 0.5 ms

SM_TMOD_IND_DIAG: Test Mode Fault Flag Indicator Check

The safety mechanism SM_TMOD_IND_DIAG diagnoses the TMODCHK flag for latent faults.

To ensure that the TMODCHK diagnostic bit is not stuck, the BMS controller takes the following steps:

1. Send the CLRFLAG command to clear any pending fault flag bit, if any;
2. Send the WRCFGA command to set the FLAG_D[7] configuration bit to force the TMODCHK bit;
3. Send the RDSTATC command to confirm that the TMODCHK flag is set;
4. Send the WRCFGA command to clear the FLAG_D[7] configuration bit again for normal operation;

5. Send the CLRFLAG command to clear the FLAG_D[7] bit for normal operation;
6. Send the RDSTATC command to confirm that the TMODCHK flag is clear.

Table 28. SM_TMOD_IND_DIAG Summary

System Integrator's Responsibilities	SM Facts
BMC to execute the above sequence.	Target: Multi point Type: On demand Impacts: all FSRs Flags: n/a Commands: CLRFLAG, WRCFGA, RDSTATC Variables: n/a Thresholds: n/a

SM_SLEEP_IND: Sleep Mode Indicator

The safety mechanism SM_SLEEP_IND reports that the device has cycled through the low-power SLEEP state, whether intentionally or unintentionally. It also confirms that the device has been properly reset after power up.

The SLEEP bit in the Status Register Group C indicates that the device has entered the sleep state, either because of a timeout condition or because of the SRST command. The SLEEP bit is also set to 1 when the device powers up. This safety mechanism has two functions: It confirms the correct power-on reset sequencing and it also reports that the device was in sleep state. When the SLEEP bit is set, the ADCs have stopped operation and the configuration bits lost their former state. The device needs to be re-initialized.

If set to 1, the SLEEP flag must be cleared by the CLRFLAG command for subsequent sleep mode entry monitoring.

Table 29. SM_SLEEP_IND Summary

System Integrator's Responsibilities	SM Facts
BMC to confirm SLEEP = 1 after reset and to clear it afterward; BMC to confirm SLEEP = 0 periodically and to re-initialize device if SLEEP = 1.	Target: Common cause Type: Periodic Impacts: all FSRs Flags: SLEEP Commands: RDSTATC, CLRFLAG Variables: n/a Thresholds: n/a Execution time: 0.5 ms

SM_NVM_ECC: Single bit error correction, multi bit error detection on protected array

The safety mechanism SM_NVM_ECC detects when the NVM or latch arrays are set to all 0s or all 1s.

The ADBMS6830 devices feature two separate sets of non-volatile memory arrays (NVM) to support the redundancy concepts. The content of the NVMs is copied into separate latch arrays when the part transitions from sleep to standby mode. The two latch arrays are protected by an Error Correction Code (ECC). Two separate ECC controllers implement a Single Error Correction, Dual Error Detection scheme with combinatorial logic. Since single-bit errors are automatically corrected by hardware, the safety concept is not concerned about the CED and SED flag bits. However, the multi-bit error flags matter. Bit CMED flags multi-bit errors in the main array and SMED the ones for the redundant array.

Where the event was triggered by a soft error, cycling through sleep state may resolve the issue.

Table 30. SM_NVM_ECC Summary

System Integrator's Responsibilities	SM Facts
BMC to check periodically whether the CMED and SMED flags are both set.	Target: Common cause Type: Periodic Impacts: all FSRs Flags: CMED, SMED Commands: RDSTATC Variables: n/a Thresholds: n/a Execution time: 0.5 ms

SM_NVM_ECC_DIAG: ECC Latent Fault Check

The safety mechanism SM_NVM_ECC_DIAG diagnoses the ECC logic for latent faults.

The ECC logic can be checked for correct functionality by toggling one and two bits in the checksum.

The BMS controller takes the following steps. The safety mechanism is only concerned about testing the multi-bit error flags CMED and SMED. Step 5 recommends clearance also of the CED and SED flags, because step 2 may cause them to be set.

1. Send the CLRFLAG command to clear any pending fault flag bit, if any;
2. Send the WRCFGA command to set the FLAG_D[6:5] configuration bits to 11 to induce a dual bit error into the ECC logic;
3. Send the RDSTATC command to confirm that the CMED and SMED bits are both set;
4. Send the WRCFGA command to clear the FLAG_D[6:5] configuration bits for normal operation;
5. Send the CLRFLAG command to clear the CED, SED, CMED and SMED bits;
6. Send the RDSTATC command to confirm that CMED and SMED bits are both clear.

Table 31. SM_NVM_ECC_DIAG Summary

System Integrator's Responsibilities	SM Facts
System integrator to ensure one transition from sleep mode to standby mode at least once ever multi-point detection time interval; BMC to execute the above sequence.	Target: Multi point Type: On demand Impacts: all FSRs Flags: n/a Commands: CLRFLAG, WRCFGA, RDSTATC Variables: n/a Thresholds: n/a

INTERFACE PROTECTION

SM_SPI_PEC: SPI Packet Error Code

The safety mechanism SM_SPI_PEC detects bit and frame errors in SPI frames. If the system integrator complements it with frame counter SM_SPI_CNT and BMS timeout detection, a full end-to-end protection can be realized. Timely response of the device is verifiable by proper PEC and CCNT values of the end of the received frames.

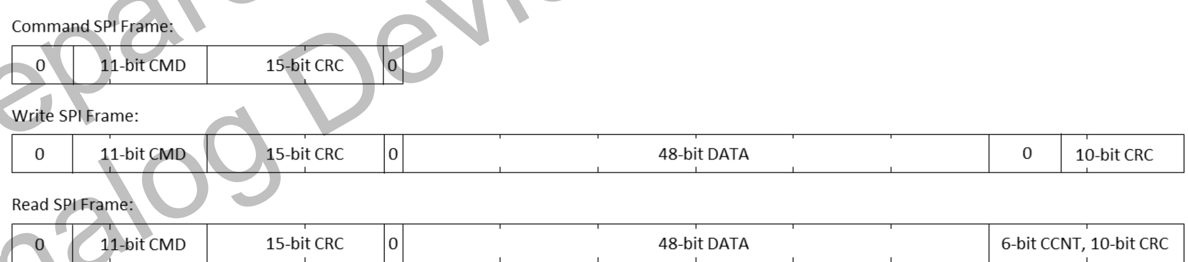

Figure 12. SPI Frames

Figure 12 illustrates the SPI frames that are relevant for the safety concept. Command SPI Frames are 32 bit long frames that are used to trigger the device. The read and the write frames transport 48 bits of payload data from and to the device.

The Command PEC (packet error code) is a 15-bit cyclic redundancy check (CRC) value calculated for all 16 bits of a command, using the following characteristic polynomial:

$x^{15}+x^{14}+x^{10}+x^{8}+x^{7}+x^{4}+x^{3}+1$ (0xC599, seed 0x0010)

This is the same as the CAN polynomial. It protects 48-bit data packets with a Hamming distance of 6. The BMS controller must send a 15-bit CRC value that represents preceding 11-bit command code. The ADBMS6830 calculates its own CRC value for a received command and compares it against the received CRC value following the command code. If the Command PEC does not match, the device does not respond to the command.

The 48 bits of payload data contained in SPI write and read frames are protected by the Data PEC. It protects payload data and command count with with a Hamming distance of 4, using a 10-bit CRC with the following characteristic polynomial:

$x^{10}+x^{7}+x^{3}+x^{2}+x+1$ (0x48F, seed 0x010)

In case of SPI read frames, the ADBMS6830 appends the calculated Data PEC to the transmitted payload data. The BMS controller must confirm that the Data PEC value from the ADBMS6830 matches its own calculation on the data returned from the ADBMS6830. The received Data PEC might be incorrect because the ADBMS6830 was not responding at all. In case of SPI write frames with payload data, the ADBMS6830 verifies also the received Data PEC. If if does not match the calculated value, the ADBMS6830 does not accept the data received. Only when it receives SPI write frames with correct Command PEC and, if the SPI frame contains payload data, with correct Data PEC, the ADBMS6830 increments the command counter to acknowledge the reception see (SM_SPI_CNT).

The read commands RDCSALL, RDACSALL and RDASALL return more bits than the used polynomial can protect with the targeted Hamming distance and are therefore not recommended for safety-relevant purposes. Refer to the ADBMS6830 data sheet for details on how to calculate the PEC.

Table 32. SM_SPI_PEC Summary

System Integrator's Responsibilities	SM Facts
BMC to append PEC checksum to SPI write frames; BMC to check PEC checksum on SPI read frames. BMC to avoid RDCSALL, RDACSALL and RDASALL read commands.	Target: Single Point Type: Continuous Impacts: all FSRs Flags: n/a Variables: n/a Thresholds: n/a

SM_SPI_CNT: SPI Frame Counter

The safety mechanism SM_SPI_CNT detects missing SPI frames and also frames, which are unintentionally transmitted twice. Timely response of the device is verifiable by proper PEC and CCNT values at the end of the frames received by the BMC.

The ADBMS6830 counts the SPI write frames (with and without payload data) received with a 6 bit wide command counter. The counter does not increment for SPI write frames which have incorrect Command PEC or Data PEC. Neither, it increments for SPI read frames. The actual count value is appended to the payload data of every SPI read frame that the device transmits. It is protected by the Data PEC. Table 33 shows how the 6-bit command count value and the 10-bit Data PEC are formatted.

The command count is the indicator whether ADBMS6830 has received SPI frames with incorrect Command or Data PEC. The BMS controller needs to confirm the expected count value after having completed a sequence of SPI commands. The command counter functions as a frame counter for write frames. The correct value confirms that no SPI write frame got lost and none was received by error. SPI frames do not cause any read side effects on the device.If a read frame is lost, the ADBMS6830 does not return any data. Consequently, the BMS controller will receive an unexpected Data PEC as in SM_SPI_PEC.

Table 33. Read Data CCNT and PEC Format

Name	RD/WR	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
PEC0	RD/WR	CCNT[5]	CCNT[4]	CCNT[3]	CCNT[2]	CCNT[1]	CCNT[0]	PEC[9]	PEC[8]
PEC1	RD/WR	PEC[7]	PEC[6]	PEC[5]	PEC[4]	PEC[3]	PEC[2]	PEC[1]	PEC[0]

The command counter is cleared at reset and during sleep state. The BMS controller can clear it with the RSTCC command. When the command counter increments past the maximum value of 63, it rolls over to a value of 1, not 0. The 0 value is reserved to those specific reset cases. Refer to the ADBMS6830 datasheet for more details on the counting rules. The command counter increments when ADBMS6830 receives valid and known commands that are marked with the “INC” tag in the Command Codes table of the datasheet.

Table 34. SM_SPI_CNT Summary

System Integrator's Responsibilities	SM Facts
BMC to initialize the command count value; BMC to check the command counter continuously.	Target: Single Point Type: Continuous Impacts: all FSRs Commands: RSTCC Flags: Variables: n/a Thresholds: n/a

SM_SPI_RED: Redundant SPI Controller

The safety mechanism SM_SPI_RED detects faults related to transmit and receive operation of the on-chip SPI controllers. It bridges from the SPI communication protection of SM_SPI_PEC and SM_SPI_CNT, to the on-chip redundancy concepts, as in SM_VCELL_RED, SM_VGPIO_RED and SM_IIR_RED.

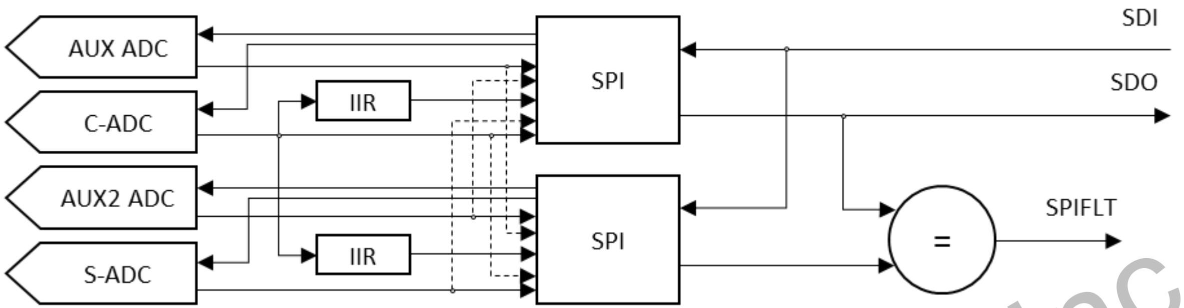

Figure 13. SM_SPI_RED and SM_IIR_RED Concept

The on-chip SPI controller is implemented redundantly, where different concepts apply:

- One SPI controller instance controls the C-ADCs and the AUXADC, the other controls the S-ADC and the AUX2ADC, extending the SM_VCELL_RED and SM_VGPIO_RED concepts;
- Result and status registers are read back by both the SPI controller instances, one after the other. A comparator compares the two resulting output bits streams and reports an SPIFLT if the two mismatch;
- Since the IIR and CAVG8 result registers are implemented redundantly themselves, each SPI instance reads only one result. The SPIFLT flag reports also the SM_IIR_RED mismatches;
- The SM_SPI_PEC and SM_SPI_CNT mechanisms are operated by both the SPI controllers and are therefore redundant as well.

If set to 1, the SPIFLT flag must be cleared by the CLRFLAG command for subsequent SPI output data monitoring.

Table 35. SM_SPI_RED Summary

System Integrator's Responsibilities	SM Facts
BMC to periodically inspect the SPIFLT flag.	Target: Single Point Type: Continuous Impacts: all FSRs Flags: SPIFLT Commands: RDSTATC Variables: n/a Thresholds: n/a Execution time: 0.5 ms

SM_SPI_PEC_DIAG: PEC Algorithm Latent Check

The safety mechanism SM_SPI_PEC_DIAG diagnose SM_SPI_PEC for latent faults.

Generally, the ADBMS6830 ignores commands with incorrect communication CRC and does not increment the command counter either. The WRCOMM, WRPWMA and WRPWMB except from this rule: If their command PEC is correct, but their data PEC is incorrect, the target register is zeroed. To diagnose SM_SPI_PEC for latent faults, command packets with single or multi-bit PEC errors are sent, once for the command PEC and once for the data PEC. The command PEC operation is the same whether the SPI write frames have payload data or not. The data PEC check applies only to SPI write frames with payload data. The command counter tells whether the command has been rejected. The data is read back to confirm that the destination register has been updated or protected as expected. A final SPIFLT check confirms that both the redundant SPI controllers counted the commands the same way during the sequence.

1. Send the RSTCC command to clear the command counter;
2. Send the MUTE command with correct command PEC;
3. Send the WRPWMA command with correct command PEC and correct data PEC to alter the PWM Register Group with any new non-zero data;
4. Send the RDPWMA command to confirm that the data from step 3 and that the command counter has incremented to 0x02;
5. Send the WRPWMA command with incorrect command PEC and correct data PEC with any new non-zero data;

6. Send the RDPWMA command to confirm that the data from step 3 and the 0x02 command count value have not changed;
7. Send the WRPWMA command with correct command PEC and incorrect data PEC with any new non-zero data;
8. Send the RDPWMA command to confirm that the returned data is zero and that the 0x02 command count value has not changed;
9. Send the RDSTATC command to confirm that the SPIFLT flag is clear;
10. Optionally, send the WRPWMA command, if the application requires non-zero value different than the reset value;
11. Optionally, send the UNMUTE command, if the application requires.

Table 36. SM_SPI_PEC_DIAG Summary

System Integrator's Responsibilities	SM Facts
BMC to execute the above sequence.	Target: Multi point Type: On demand Impacts: all FSRs Flags: n/a Commands: WRPWMB, Read Cmd on register or command counter Variables: n/a Thresholds: n/a

SM_SPI_RED_DIAG: SPI Compare Logic and Fault Flag Indicator Check

The safety mechanism SM_SPI_RED_DIAG diagnose the SPI compare logic and the SPIFLT flag for latent faults.

The SPI compare logic and the SPIFLT flag diagnostic bit can be checked that is not stuck by issuing an RDSTATC command with bit ERR set to 1.

The BMS controller must take following steps:

1. Send the CLRFLAG command to clear any pending fault flag bit, if any;
2. Send RDSTATC command to confirm that the SPIFLT bit is clear;
3. Send RDSTATC command with the ERR parameter bit set to confirm that the SPIFLT bit is set;
4. Send RDSTATC command with the ERR parameter bit clear to confirm that the SPIFLT bit is clear.

Table 37. SM_SPI_RED_DIAG Summary

System Integrator's Responsibilities	SM Facts
BMC to execute the above sequence.	Target: Multi point Type: On demand Impacts: all FSRs Flags: n/a Commands: CLRFLAG, RDSTATC Variables: n/a Thresholds: n/a

SM_FREEZE_SEQ: No Clock and Freeze Detector

Measurement results gained during continuous operation need to be "frozen" for proper synchronization with the SPI operation. If stuck frozen active, then the updated conversion results will not progress to the SPI interface. If stuck frozen inactive, then the data sent by the SPI read frames may not be consistent with itself and the snap operation may not be reliable. The safety mechanism SM_FREEZE_SEQ detects if measurement results gained during continuous operation are stuck frozen, whether active or inactive. An alternative version detects the stuck frozen active in the single-shot operation case, because a stuck frozen inactive is not a concern there. The mechanism also detects the failure mode "no clock" of the main oscillator.

The freeze and snap mechanisms are common to all channels results, including filter and averaging results. Therefore, only one channel needs to be inspected by the below sequence, that should be executed right after the RDSTATC issued because of SM_VCELL_RED. The sequence assumes that the C-ADCs are operating in continuous mode triggered by an ADCV(CONT=1) command more than one millisecond ago.

1. Send the SNAP command;
2. Send the CLRCELL command to clear the Cell Voltage Registers;
3. Send the CLRFLAG command to clear all 16 CSxFLT bits;
4. Wait for 1 ms;

5. Send the RDCVA command to confirm that all registers return 0x8000 values;
6. Send the UNSNAP command;
7. Send the RDCVA command to confirm that all returned register values are in range as SM_VCELL_OOR expects.

Since step 3 clears the CSxFLT flag bits, the status of these bits needs to be read beforehand. The wait period in step 4 can be filled with other SPI commands that are not impacted by CLRCELL and SNAP.

The CLRCELL command is destructive and clears C-ADC and CAVG8 results during on-going operation. However, the sequence does not disturb the IIR filter operation any.

Applications that operate the C-ADC in single-shot mode always, can use the following simplified version of the sequence:

1. Send the ADCV(CONT=0) command;
2. Send the CLRCELL command;
3. Send the RDCVA command to confirm that all registers return 0x8000 values;
4. Wait for the ADCV conversion to complete (1 ms from step 1);
5. Send the RDCVA command to confirm that all returned register values are in range as SM_VCELL_OOR expects.

The steps 1 to 3 need to complete within one millisecond. Step 3 contributes to the latent fault metric only and needs to execute only once per multi-point fault detection interval. The other steps are mandatory periodically.

Table 38. SM_FREEZE_SEQ Summary

System Integrator's Responsibilities	SM Facts
BMC to periodically execute the above sequence.	Target: Single point Type: Periodic Impacts: all FSRs Flags: n/a Commands: ADCV,SNAP, CLRCELL, UNSNAP, RDCVA Variables: n/a Thresholds: n/a Execution time: around 2 ms

SM_CFG_RED: Redundant Configuration Registers

The safety mechanism SM_CFG_RED detects faults related to the configuration registers.

Note: Faults in configuration registers will be detected by the effect of different configuration on main and redundant path.

Internally, the following configuration bits are implemented redundantly:

- REFON: Reference Powered Up control bits
- CTH: C vs S Comparison Voltage Threshold
- SOAK feature related bits
- FC: IIR Filter Parameter

The redundant copies are not visible to the SPI interface. Errors in the configuration bits will result in different main and redundant measured values. The voltage references are not powered up if one of the redundant REFUP configuration bits fails to be set.

Table 39. SM_CFG_RED Summary

System Integrator's Responsibilities	SM Facts
none	Target: Single Point Type: Continuous Impacts: all FSRs Flags: Commands: Variables: n/a Thresholds: n/a Execution time:

ASSUMPTIONS OF USE

The assumptions of use (AoU) can be seen as the safety requirements for the user who is using the ADBMS6830 devices in safety applications. The safety concept and the safety analysis developed by Analog Devices rely on these assumptions being fulfilled by the system integrator. This section does not include assumptions made on the applications for correct operation of the device outside of the functional safety context.

Every AoU is tagged with a "Type" tag.

- "Mandatory" means that the safety concept will not work without the assumption being true.
- "Recommended" means that the safety concept may still be valid if the assumed behavior is not implemented. However, there will be a hit to diagnostic coverage metrics and the ASIL targets may no longer be met. Judgement on the qualitative and quantitative impact is with the system integrator.
- "Supportive" means that the assumption made during the development of the safety concept has not got any credit in the safety analysis of the ADBMS6830 device. Still, it may support the system-level safety concept.

Each ADBMS6830 device will be used to monitor in series connected battery cells. The maximum number of monitored cells depends for each device. The cells are connected between C-pins via a simple 1-pole RC filter connected to every C-pin. The cells are also connected between corresponding S-pins via a simple 1-pole RC filter. See Figure 3 for details.

Each ADBMS6830 will be used to monitor battery cells temperatures. The cell temperature measurements will be made using two thermistors and resistor combinations connected to two GPIOs. The measured thermistors voltage range will be from 0.1V to 3.0V. See Figure 9. ADBMS6830 use conditions must comply with the following:

GENERAL ASSUMPTIONS

Table 40. Compliance

ID	Description
AoU01	The system is designed to maintain the IC temperature in compliance with ADBMS6830 device datasheet. Type: Mandatory
AoU02	The ADBMS6830 device is operated within all specified voltage and current ratings defined in ADBMS6830 datasheet. Type: Mandatory
AoU20	Only ADBMS6830 generics that are labeled with 'WFS' in their part name are used in automotive safety applications. Rationale: Only 'W' generics pass the automotive production flow. Only 'FS' generics are controlled by the full safety life cycle. Type: Mandatory
AoU21	The system integrator ensures that abnormal device behavior as documented in the ADBMS6830 Errata Sheet is not violating any of the safety goals. Type: Mandatory
AoU22	The system integrator ensures that the usage of the ADBMS6830 device is compliant with this Safety Manual. Type: Mandatory
AoU23	Any safety information provided by the Safety Manual is applied solely to the scope of applications as addressed by the ISO 26262-2018 series of standards. Rationale: No other standards outside of the automotive application space have been considered during development. Type: Recommended

Table 41. General Safety Integration

ID	Description
AoU24	The system integrator validates whether the assumed safety requirements are consistent with the needs of the final product. In case of a delta, the system integrator verifies whether the SEooC is suitable for use. Rationale: No analysis is available, whether the element mitigates requirements not known during development time. Type: Mandatory
AoU03	Circuit configuration and external components for the ADBMS6830 comply with ADI recommendations in the ADBMS6830 datasheet. Rationale: While other configurations, components, and component characteristics may be suitable, ADI is not in the position to verify or quantify the safeness. Type: Recommended
AoU07	The system integrator defines the fault detection time interval (FDTI) and the multi-point fault detection time interval (MPFDTI) at the system level and ensures their compliance with the safety concept of this safety manual. Rationale: The Fault Detection Time specified along with the assumed safety requirements have been used as guidance for the development of the device. The device may support faster detection times, especially when only a subset of the features is used in the application. Type: Mandatory
AoU25	The system integrator addresses all duties listed under "System Integrator's Responsibilities" for all safety mechanisms. Type: Mandatory
AoU26	The system integrator ensures that all Safety Mechanisms, that target single-point, common-cause or cascading faults and are classified as "Continuous" are applied whenever consuming the data. Type: Recommended
AoU27	The system integrator ensures that all Safety Mechanisms, that target single-point, common-cause or cascading faults and are classified as "Periodic" are executed at least once per the Fault Detection Time Interval required by the application. Implementation Guide: A diagnostic test time interval (DTTI) faster than the FDTI can further increase the robustness against transient faults. Type: Recommended
AoU28	The system integrator ensures that all Safety Mechanisms, that target multi-point faults are executed at least once per multi-point fault detection time interval defined at system level. Implementation Guide: Ideally, the mechanisms execute at start-up time every driving cycle and every charging cycle. Type: Recommended

APPLICATION-SPECIFIC ASSUMPTIONS

Table 42. Specific SEooC Assumptions

ID	Description
AoU00	The FTTI is define at system level and the FTTI is assumed to be no less than 100 ms. Type: Mandatory
AoU04	The ADBMS6830 devices should be connected to a system controller capable of implementing all specified BMS control functions with the appropriate ASIL, either directly through the SPI interface or indirectly through an isoSPI daisy chain. Type: Mandatory
AoU13	The ADBMS6830 cell monitors are sensor devices and do not carry any fault reaction or fail operational responsibilities. If powered down or not operational, the devices stop responding to commands from the BMC and it is up to the BMC to manages system safety.

	Type: Mandatory
AoU08	The BMS controller resets the Command Counter after exiting the LPCM mode. Rationale: Start with a clean state of the counter. Type: Mandatory
AoU10	The system assumes that any single measurement can be corrupted due to low-probability transient faults, and is designed to tolerate such events. Type: Mandatory
AoU16	Where wires between the battery cells and the ADBMS6830 serve multiple purposes, the system integrator controls by system measures that voltage drop cross the wire resistance is not introducing common-cause failures. Rationale: Cables and connectors between the battery cells and the cell monitor may have non-negligible resistance that may even vary over life time. The supply current of the ADBMS6830 may cause a voltage drop cross the connections that could lead to an undetected error in cell voltage and temperature measurement. Especially when the NTC resistance values are low because of high temperatures, the voltage drop on common ground returns may need to be considered. See sections Open Supply Wires and Open Cell Temperature Sense Wires. Implementation Guide: Unless the connection impedance is negligible, have dedicated wires for supplying power to the V+ and V- pins of the ADBMS6830 that are separate from the cell voltage sense lines and separate from the NTC ground returns. Ensure that redundant NTCs are not using common ground return wires with significant voltage drop. Type: Mandatory
AoU17	Where external diodes are installed from cell sense wires to the V+ supply, the system integrator ensures that the diode does not introduce undetectable failure modes. Rationale: Such diodes may ensure communication of the device and the isoSPI daisy chain if the V+ supply wire between battery and device breaks, but have not been formally verified by the ADBMS6830 safety concept. See section Open Supply Wires. Implementation Guide: Options for counter measures include redundant wiring, V+ measurement with the AUX ADC and diode networks that impact C-ADC and S-ADC channels in significant asymmetrical way, and are to be verified by the system integrator. Type: Recommended

Table 43. Use Case Constraints

ID	Description
AoU11	The cell balancing functionality is not considered a safety function of the ADBMS6830 devices. Rationale: Unintended activation of the discharging function can drain the battery cells. However, due to the external series resistors even a short between adjacent cell monitoring pins can discharge a cell only with a moderate current. Cell discharging was slow and enabled the algorithm on the BMS controller to control safety over the course of a longer period, based of the conversion results of the regular cell voltage monitoring. Similarly, if the dischargers were stuck off while intended to be enabled, the algorithm can detect the unexpected response of the measured cell voltage. Type: Mandatory
AoU12	The low-power cell monitoring (LPCM) functionality is not considered a safety function of the ADBMS6830 devices. Rationale: While in LPCM mode the vehicle is neither driving nor charging. The contactors are assumed to be already open and hence the battery management system is already in a safe state. Type: Mandatory
AoU14	The master SPI/I2C functionality through the COMM port is not considered a safety function of the ADBMS6830 devices. Rationale: The ADBMS6830 provides a physical interface to local SPI or I2C devices, that can be connected through the GPIO pins. However, at no point the ADBMS6830 accesses the EEPROM by its own command. Neither, it ever interprets the content of the EEPROM directly. The ADBMS6830 behaves like a black channel. The safety concept neither ensures freedom of interference between COMM-port communication and other safety functions at runtime, nor does it plan for FDTI requirements to be met, while COMM-port communication is ongoing. Implementation Guide: Where the GPIOs are used to connect SPI or I2C devices, access them while the ADBMS6830 is not converting data. Verify their safeness by application-level means. For example, protect the content of connected a non-

	volatile memory device, such as an EEPROM, by reading the entire content at once and by protecting the data with a full end-to-end software checksum that is generated and verified by the BMC. Type: Recommended.
AoU15	The retention register functionality is not considered a safety function of the ADBMS6830 devices. Rationale: The ADBMS6830 features scratch registers that retain their content even if the VREG supply is powered off. The device hardware does not protect data storage and access with safety mechanisms. Implementation Guide: Where the retention registers are used to store safety relevant information, protect the bits with a software checksum, that is generated and verified by the remote BMC. Type: Recommended.

Earlier versions of this safety manual listed also the assumptions of use AoU05, AoU06 and AoU09. These assumptions are unchanged. However, they are now disclosed as "System Integrators' Responsibilities" in Table 13 as they belong to SM_VGPIO_RED.

SAFETY ANALYSIS

HARDWARE METRICS

Quantitative analysis of random hardware faults assumes that the ADBMS6830 is used within the scope of the stated Table 42. The system must utilize the diagnostic features of the ADBMS6830 to ensure that measurement results are not in error.

This section provides customers guidance on the approach Analog Devices used in calculating the preliminary ISO 26262 metrics of Single Point Fault Metric (SPFM), Latent Fault Metric (LFM) and Probabilistic Metric for random Hardware Failures (PMHF). This document assumes the reader is familiar with the ISO 26262 metrics in question and with the fundamentals of quantitative safety analysis. There is no intent to act as a tutorial on these topics.

In general, the approach taken was to remain conservative on all aspects of parameters which could impact the SPFM, LFM and PMHF calculations. Achieving the ASIL targets with a conservative approach provides higher confidence to the program. Further analysis may refine certain parameters to make them less conservative.

Assumed Mission Profile

This document does not intend to specify reliability data. For comparability, the quantitative calculations in this safety analysis are based on a product life time to last 15 years and the following assumed mission profile:

Table 44. Assumed Mission Profile

IC Ambient Temperature	Percentage
-40°C	6%
+23°C	20%
+50°C	65%
+100°C	8%
+105°C	1%
+43.75°C (average)	100% (total)

The calculation assumes an ontime of 40%, given that the Cell Monitors powered by the battery cells for the entire lifetime, but operate only during driving and charging cycles. Additionally, the low power cell monitoring (LPCM) feature requires the devices to wake up once in a while during parking mode.

Random Base Failure Rate

The base failure rate is the rate applied to the analysis prior to any benefit for safeness or safety mechanisms being applied. The random base failure rate is composed of permanent contribution following the IEC TR 62380 guidance.

Table 45. IEC 62380 Parameters

IEC TR 62380 Parameters	Package + Main Die	Zener Reference Die
λ_1	2.7e-3	2.2e-2
λ_2	20	3.3
λ3	$0.048*D^{1.68}$	
α_S	16	16
α_C	21.5	21.5
τ_i, τ_{on} and τ_{off} $[(\pi_t)_i]$	from mission profile	
π_I	0	0
λ_{EOS}	0	0
Power dissipation	0.223	0.0025
D	Peripheral connection packages	

For the thermal junction-to-ambient resistance a value of 25.9°C/W is used for the LQFP80 package and 20.19°C/W for the QFN72 package. This results in the base failure rate estimation as in the table below:

Table 46. IEC 62380 Base Failure Rates

IEC TR 62380 λ	ADBMS6830 LQFP80	ADBMS6830 QFN72
λ_{main_die}	11.62 FIT	11.06 FIT
λ_{zener_die}	1.97 FIT	1.97 FIT
$\lambda_{package}$	58.25 FIT	44.70 FIT
λ_{total}	71.84 FIT	57.73 FIT

It can be seen that failure rate benefits from the smaller physical dimensions of the QFN72 package significantly.

Diagnostic Coverage

Where possible, diagnostic coverage estimates were taken from ISO 26262. Even in the scenarios where these are known to be conservative they were left unchanged. In scenarios where Analog Devices had developed a safety mechanism which did not map to those detailed in the standard, expert judgement was used to estimate the expected diagnostic coverage.

In scenarios where a single failure mode was covered by multiple safety mechanisms, diagnostic coverage credit was only taken for one safety mechanism, the one with the highest coverage value. Unless the ISO 26262 standard credits the combination of safety mechanisms explicitly (such as for frame counter + checksum + timeout) no benefit was taken in the calculation for the other safety mechanisms, i.e. no attempt to add the diagnostic coverage of safety mechanisms was made.

HW Metrics Calculation

Separate calculation models have been performed for the two package variants. Each model calculates the hardware metrics for the two safety goals, represented by safety requirements FSR1 and FSR2. SPFM and LFM have been gained by FMEDA analysis. Also the PMHF was derived from FMEDA and counterpartyed by FTA.

Table 47. Hardware Metrics Summary for the ADBMS6830 LQFP80 Package

Parameter	FSR1	FSR2
Total Failure Rate, λ	71.84 FIT	71.84 FIT
Safety Related Faults, λ_{SR}	62.92 FIT	24.75 FIT
Residual Faults, λ_{RF}	0.34 FIT	0.19 FIT
Latent Faults, $\lambda_{MPF,L}$	5.99 FIT	2.22 FIT
SPFM	99.46 %	99.26 %
LFM	90.43 %	90.96 %
PMHF (IEC 62380)	0.39 FIT	0.20 FIT

Table 48. Hardware Metrics Summary for the ADBMS6830 QFN72 Package

Parameter	FSR1	FSR2
Total Failure Rate, λ	57.73 FIT	57.73 FIT
Safety Related Faults, λ_{SR}	50.65 FIT	21.24 FIT
Residual Faults, λ_{RF}	0.27 FIT	0.16 FIT
Latent Faults, $\lambda_{MPF,L}$	4.80 FIT	1.90 FIT
SPFM	99.48 %	99.26 %
LFM	90.48 %	91.00 %
PMHF (IEC 62380)	0.29 FIT	0.17 FIT

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